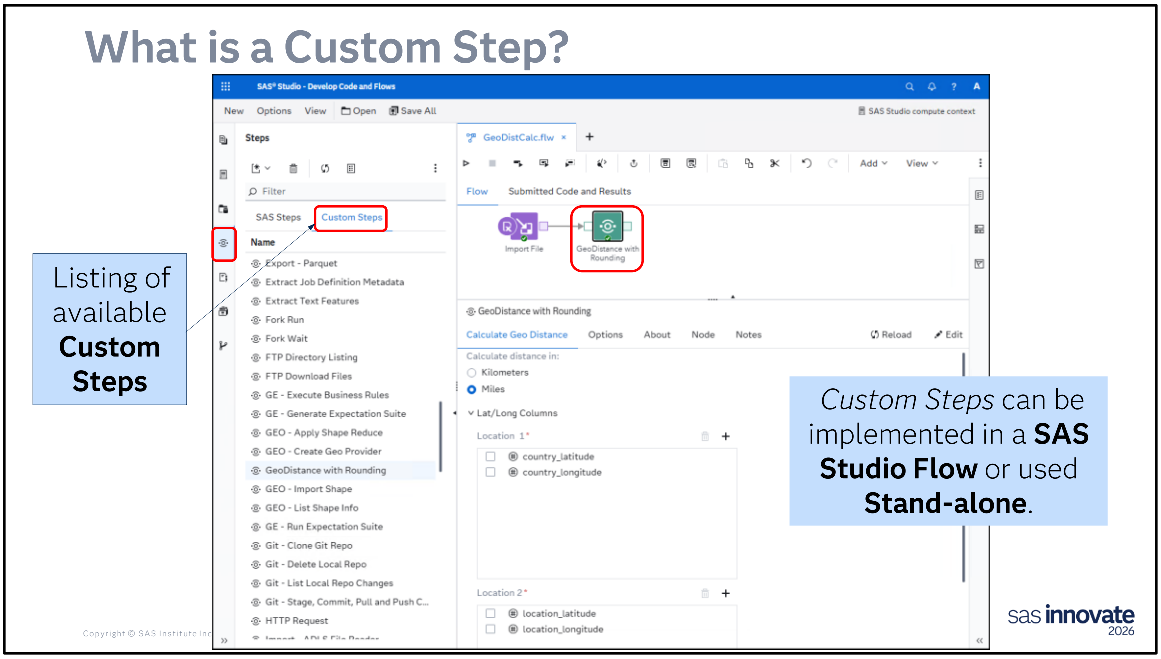
Tips & Tricks for Building a Custom Step with SAS® Viya

Hands-on Workshop

Mary Kathryn Queen
Data Engineering and Decisioning
SAS Education Content Development Group

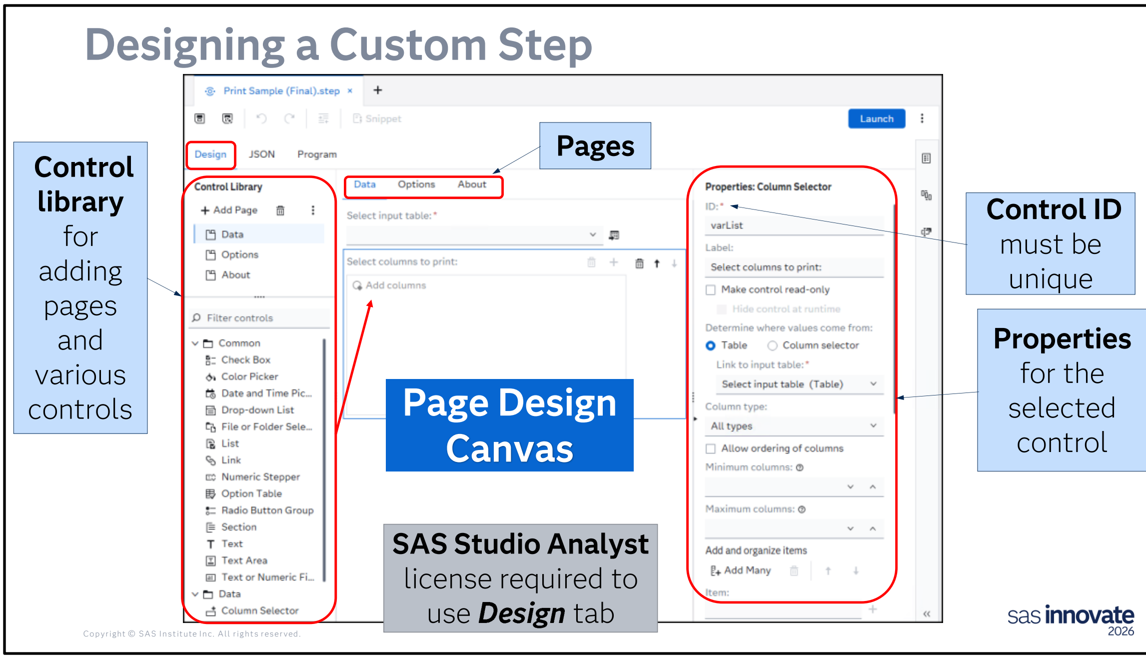
#SASInnovate


sas innovate 2026
Tips & Tricks for Building a Custom Step with SAS® Viya
Hand-on Workshop
Mary Kathryn Queen
Data Engineering and Decisioning
SAS Education Content Development Group
Copyright © SAS Institute Inc. All rights reserved.



Custom Steps enable you to create a user interface to complete a specific task. Available custom steps are listed on the **Steps** pane in *SAS Studio* on the **Custom Steps** tab.

They can then be used in multiple SAS Studio Flows or stand-alone to ensure a common, repeatable process.

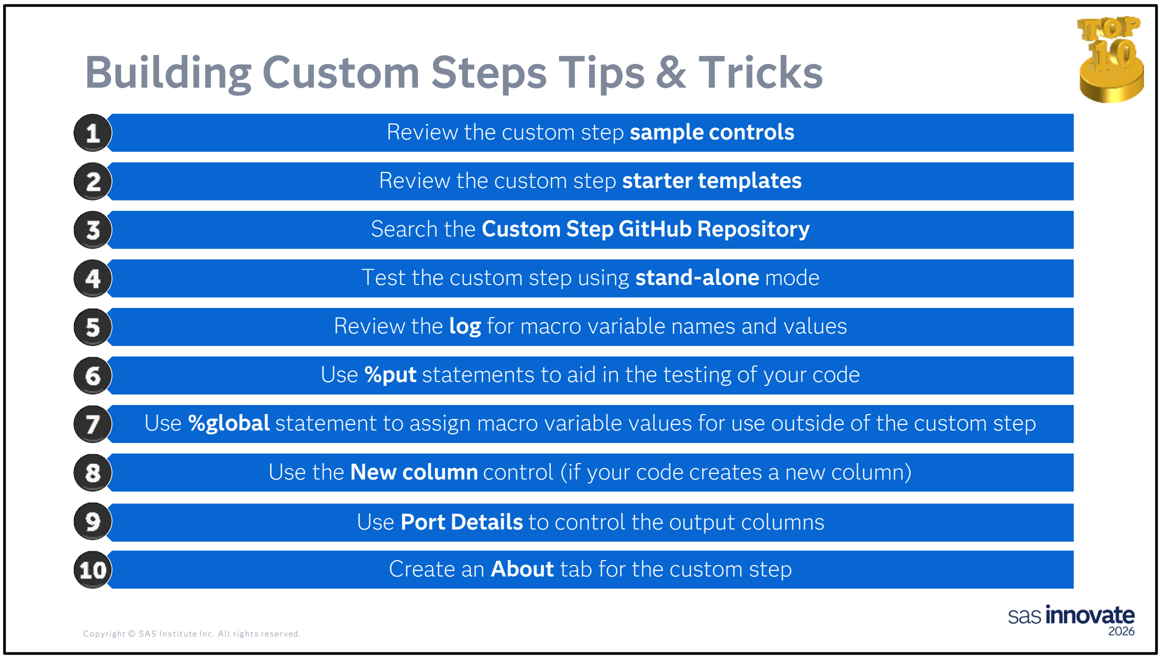


The **Design** tab is a visual interface to help you quickly build the JSON for the user interface for your custom step. This JSON is visible on the **JSON** tab. A **SAS Studio Analyst** license is required to use the **Design** tab; if you just have a **SAS Studio Basic** license, then you will need to design your custom steps using only the **JSON** tab.

On the left-hand side is the Control library for adding **pages** and **controls** to the user interface. **Pages** become tabs in the user interface for the custom step.

Each page has its own page design that you customize by adding controls to the **Page Design Canvas**.

For each page or control their **properties** are set on the right-hand side. Each control must have a unique ID within the Custom Step.



1. The **sample controls** provide an explanation for each of the control types you can use when designing a custom step.
2. The **starter templates** are example custom steps with associated program code, so you can review some real-life examples of how they were built.
3. Search the **Custom Step GitHub Repository** for any similar custom steps that you may be trying to build and download them to use as a starting point for your design. https://github.com/sassoftware/sas-studio-custom-steps
4. Use the **stand-alone** mode to quickly test your custom step.
5. Review the **log** for the macro variable names and their respective values.
6. Use **%put** statements to aid in the testing of your code. This allows you to write out values or show that you have reached a particular portion of the code.
7. Use a **%global** statement to assign macro variable values for use outside of the custom step, if needed.
8. Use the **New column** control if the code within your custom step creates a new column. You can hide this column if you don't want the

end user to change the name or you can choose to display it if you want to give the end user the opportunity to name the new column.

9. Use **Port Details** to control the output columns from the custom step. If removing columns, you also need to remove them in the code of the custom step.
10. Create an **About** tab to provide a description of the custom step and its release date information.

[SASInnovate2026/Tips-Tricks-for-Building-a-Custom-Step-in-SAS-Viya: Tips & Tricks for Building a Custom Step in SAS Viya](https://github.com/SASInnovate2026/Tips-Tricks-for-Building-a-Custom-Step-in-SAS-Viya)
https://github.com/SASInnovate2026/Tips-Tricks-for-Building-a-Custom-Step-in-SAS-Viya

Tips & Tricks for Building a Custom Step with SAS Viya

Exercise Description

In this hands-on exercise, you will review the top 10 tips and tricks for creating a custom step in SAS Studio.

Log in to SAS Viya

Open a new window in the *Google Chrome* browser and select the **SASLanding** bookmark.

- ID: **student**
- Password: **Metadata0**

Select **No** when prompted about accepting *Admin* privileges.

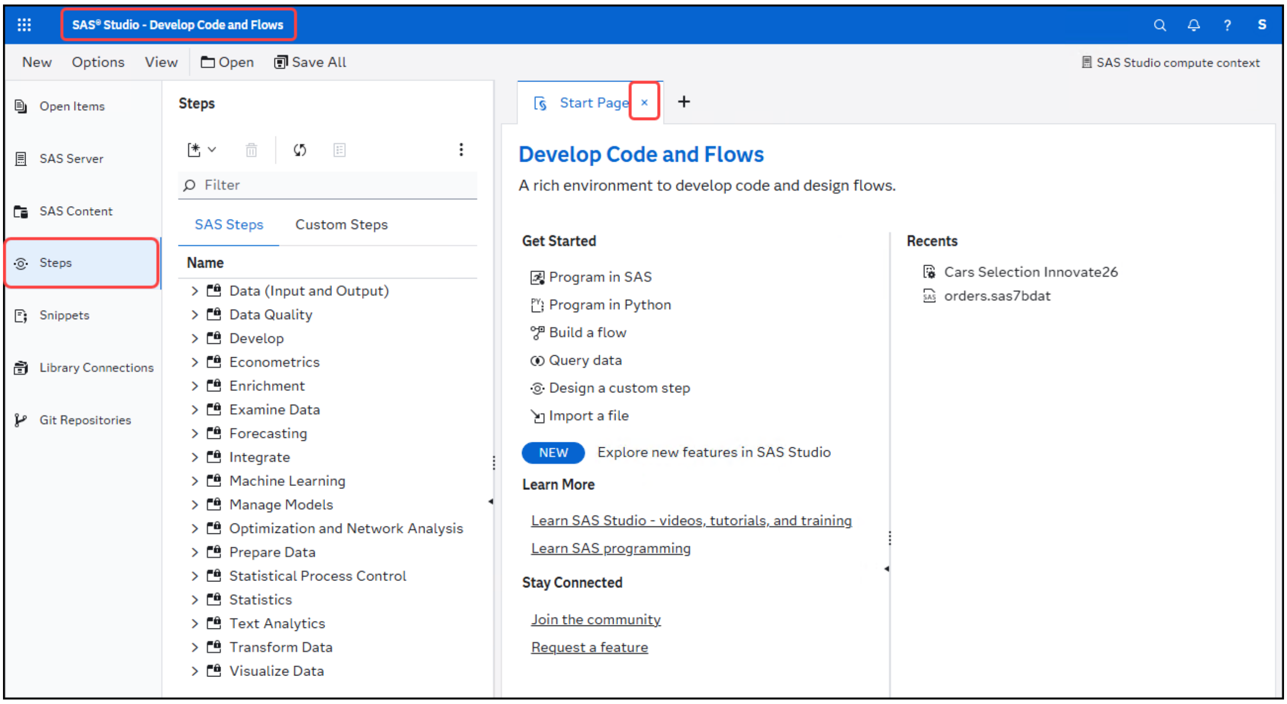
Top 10 Tips & Tricks for Building a Custom Step

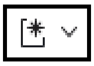
1. Review the Custom Step Sample Controls

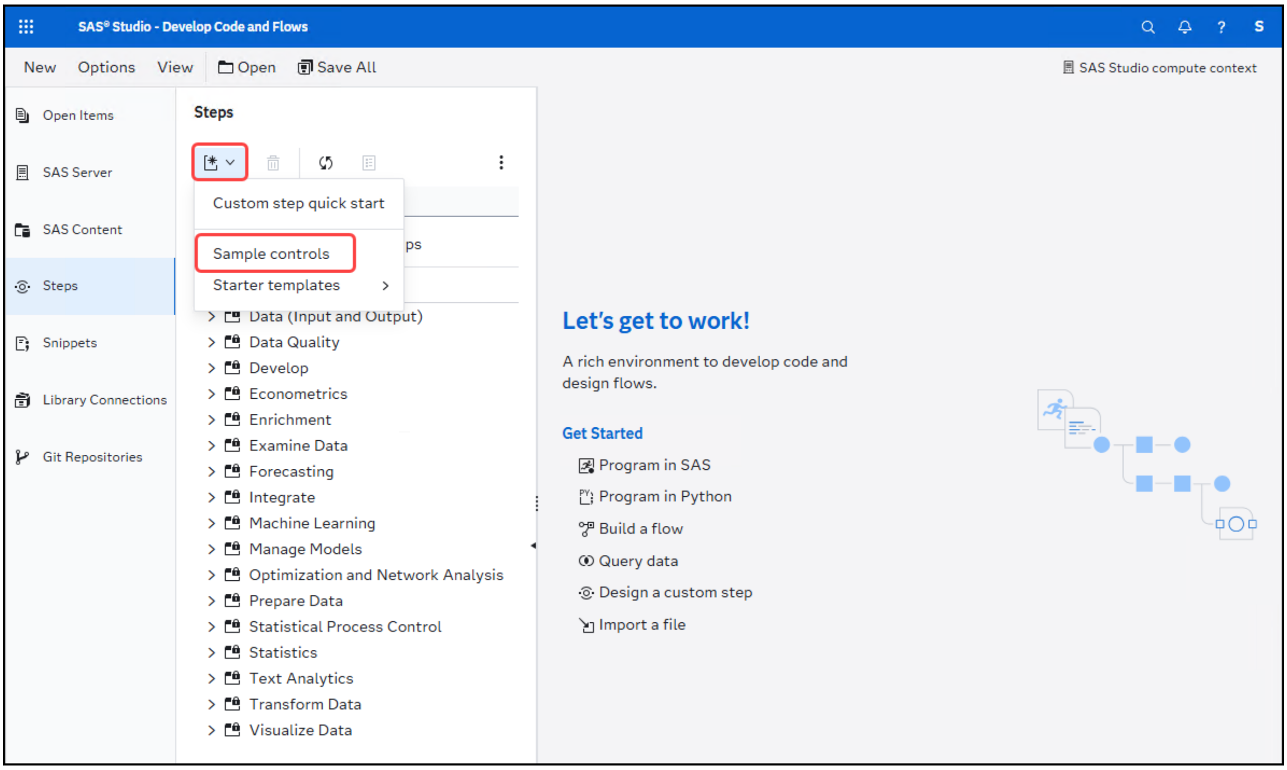
1. Select ➔ **Develop Code and Flows** to open *SAS Studio*.

2. Click **x** to close the *Start Page*, if it is open. Also, close any other open tabs, if needed.

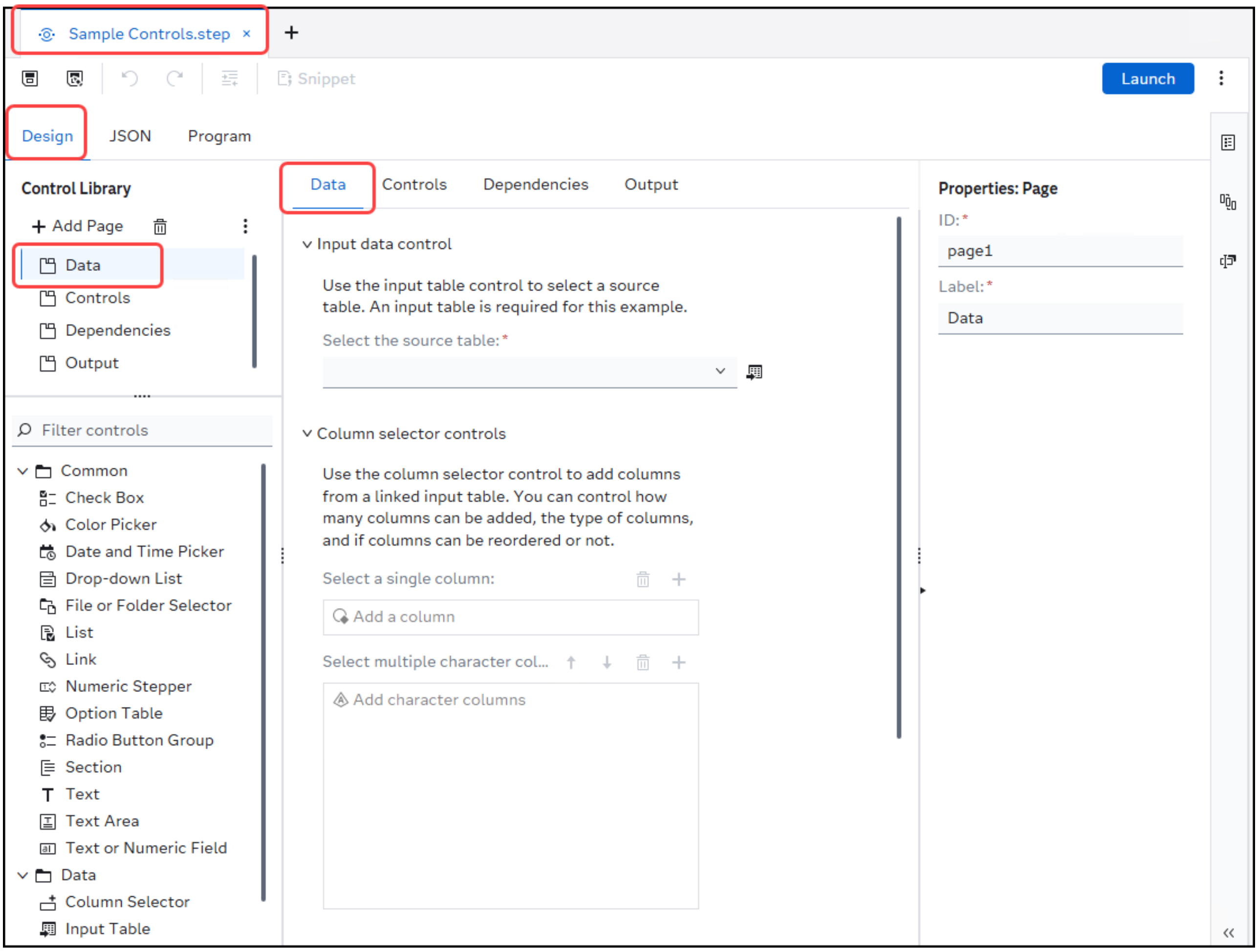
3. Select to view the **Steps** pane.



4. On the *Steps* pane, select  ➔ **Sample controls**.

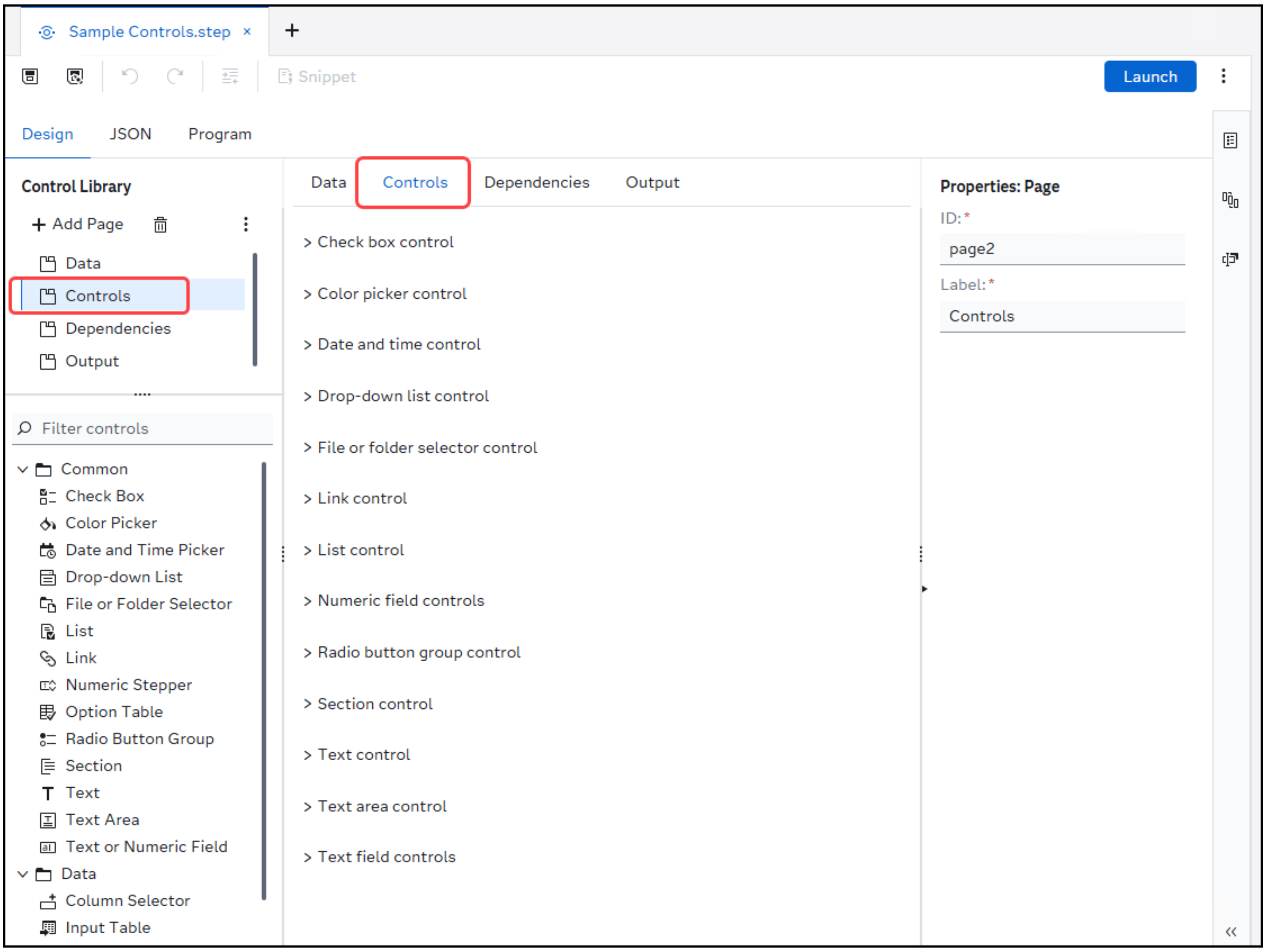


5. On the *Design* tab, select **Data** in the *Control Library* section to view the *Data* page of the *Sample Controls* custom step.



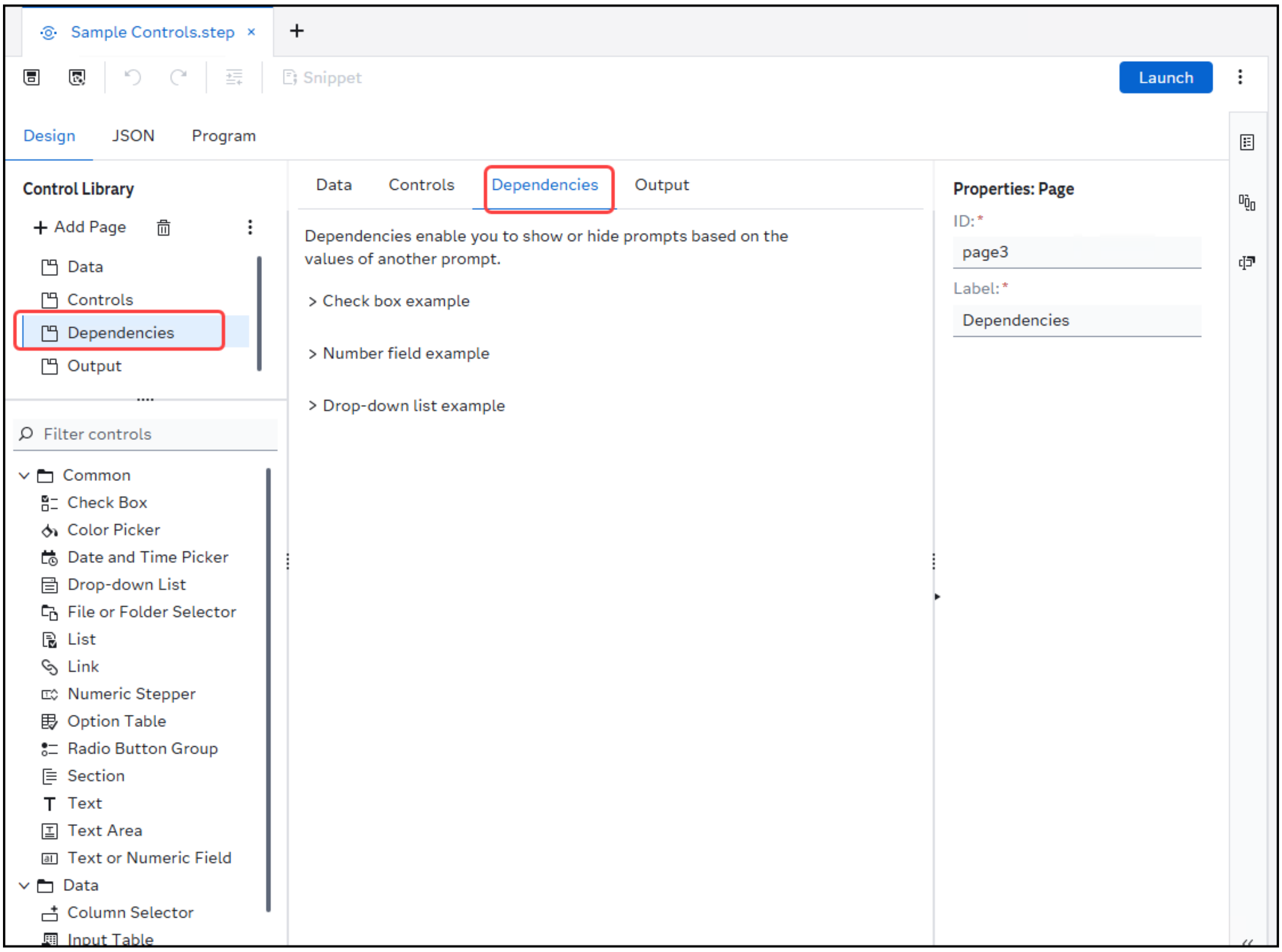
> ✎ There is an example and text explanation for each of the **Data** controls with the exception of the *Output Table* data control. There is an example and text explanation of that control on a another page.

6. Select **Controls** in the *Control Library* section to view the *Controls* page of the *Sample Controls* custom step.



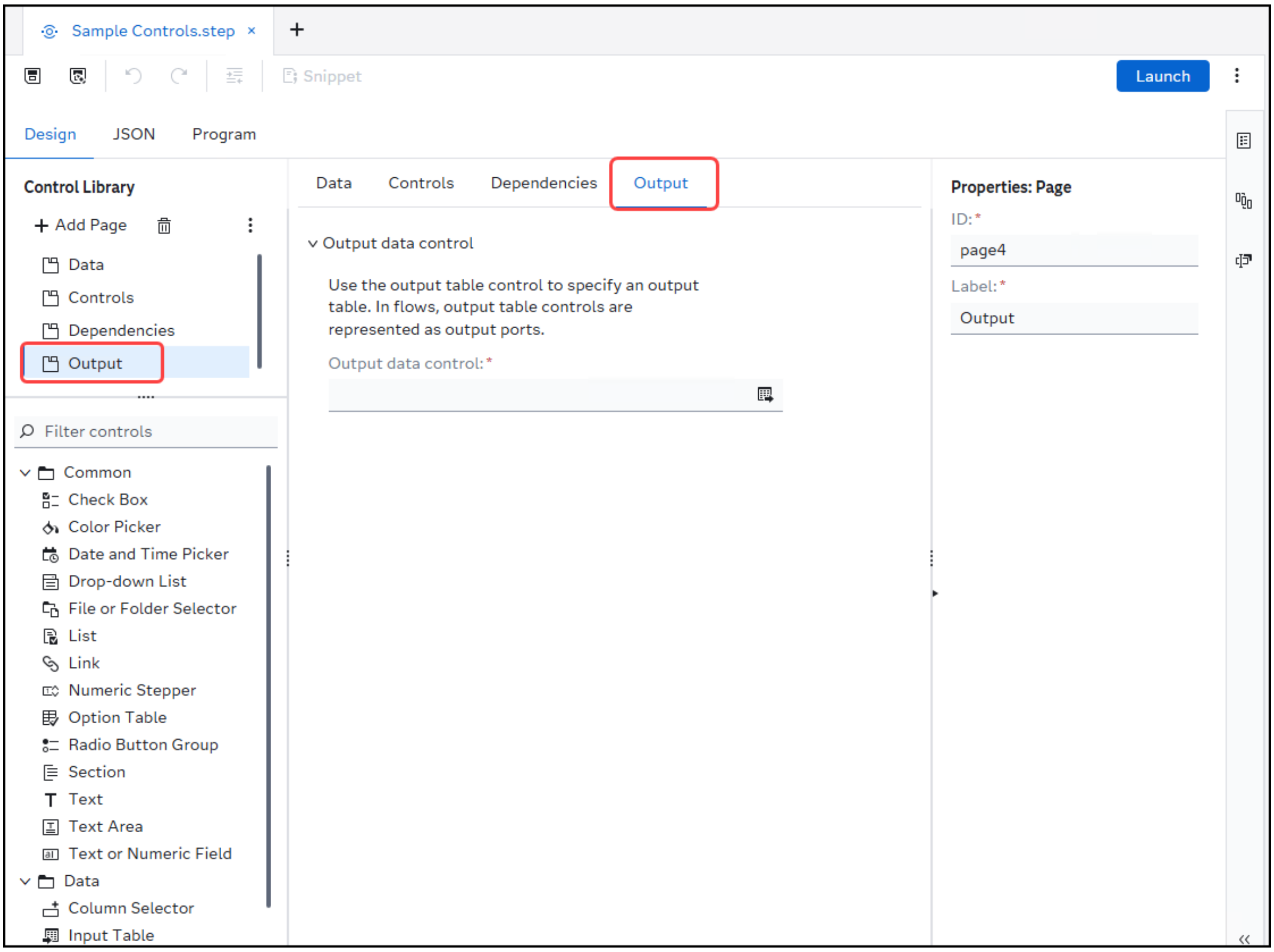
> ✎ There is an example and text explanation for each of the **Common** controls. Select **>** to expand the control to view each example and text explanation.

7. Select **Dependencies** in the *Control Library* section to view the *Dependencies* page of the *Sample Controls* custom step.



> ✎ There is an example and text explanation for dependencies with *check box*, *number field*, and *drop-down list*. Select **>** to expand the control to view each example and text explanation.

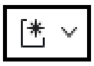
8. Select **Output** in the *Control Library* section to view the *Output* page of the *Sample Controls* custom step.

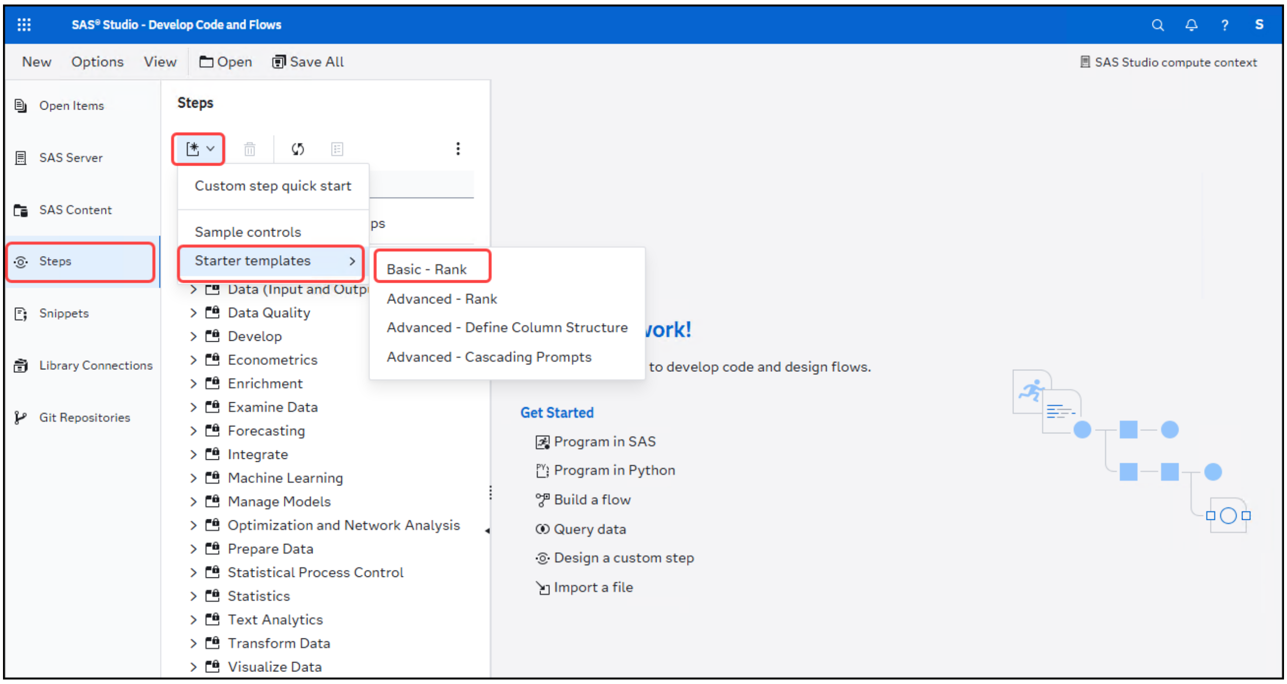


> ✎ There is an example and text explanation for **Output table** *Data* control.

9. Click **x** to close the *Sample Controls.step* tab.

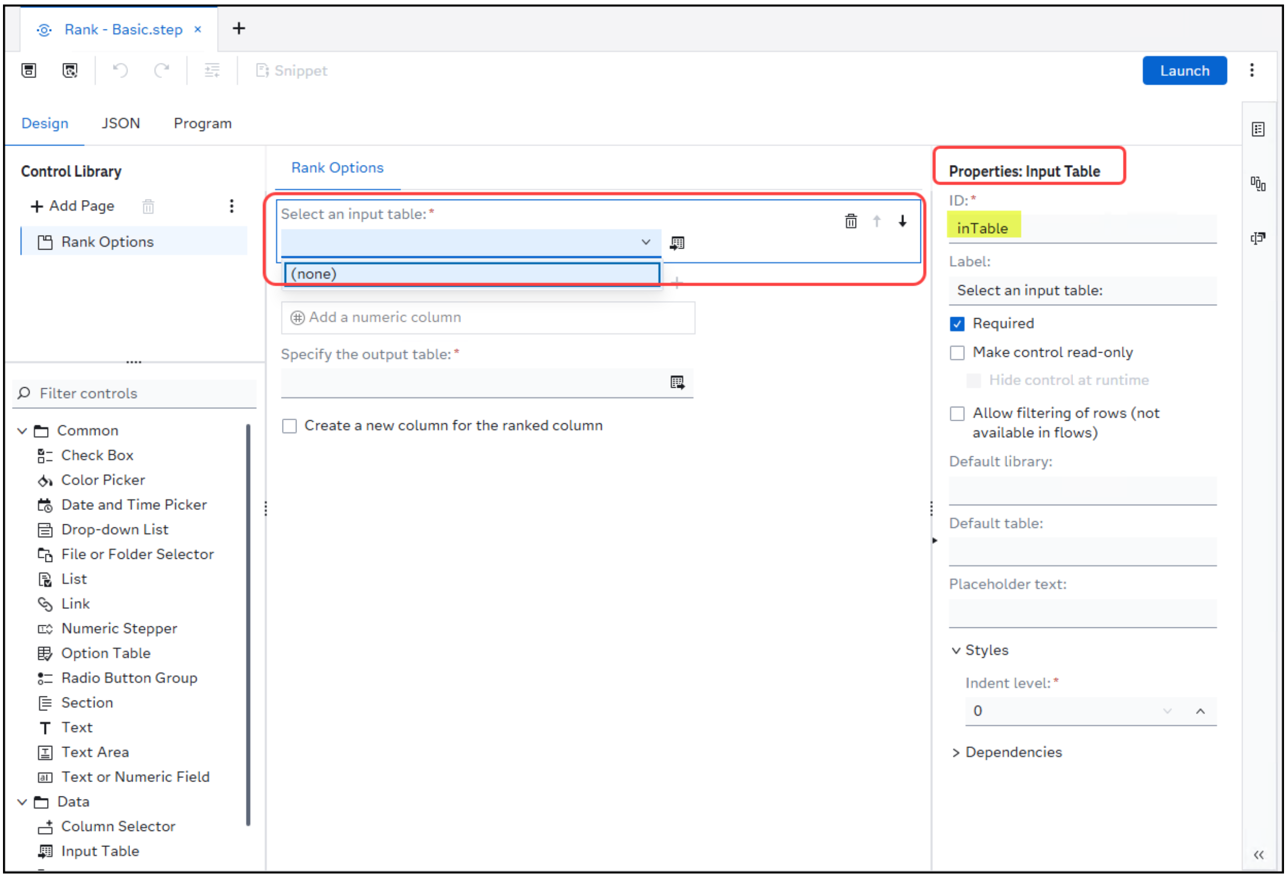
2. Review the Custom Step Starter Templates

1. On the *Steps* pane, select  **➔ Starter templates ➔ Basic - Rank**.



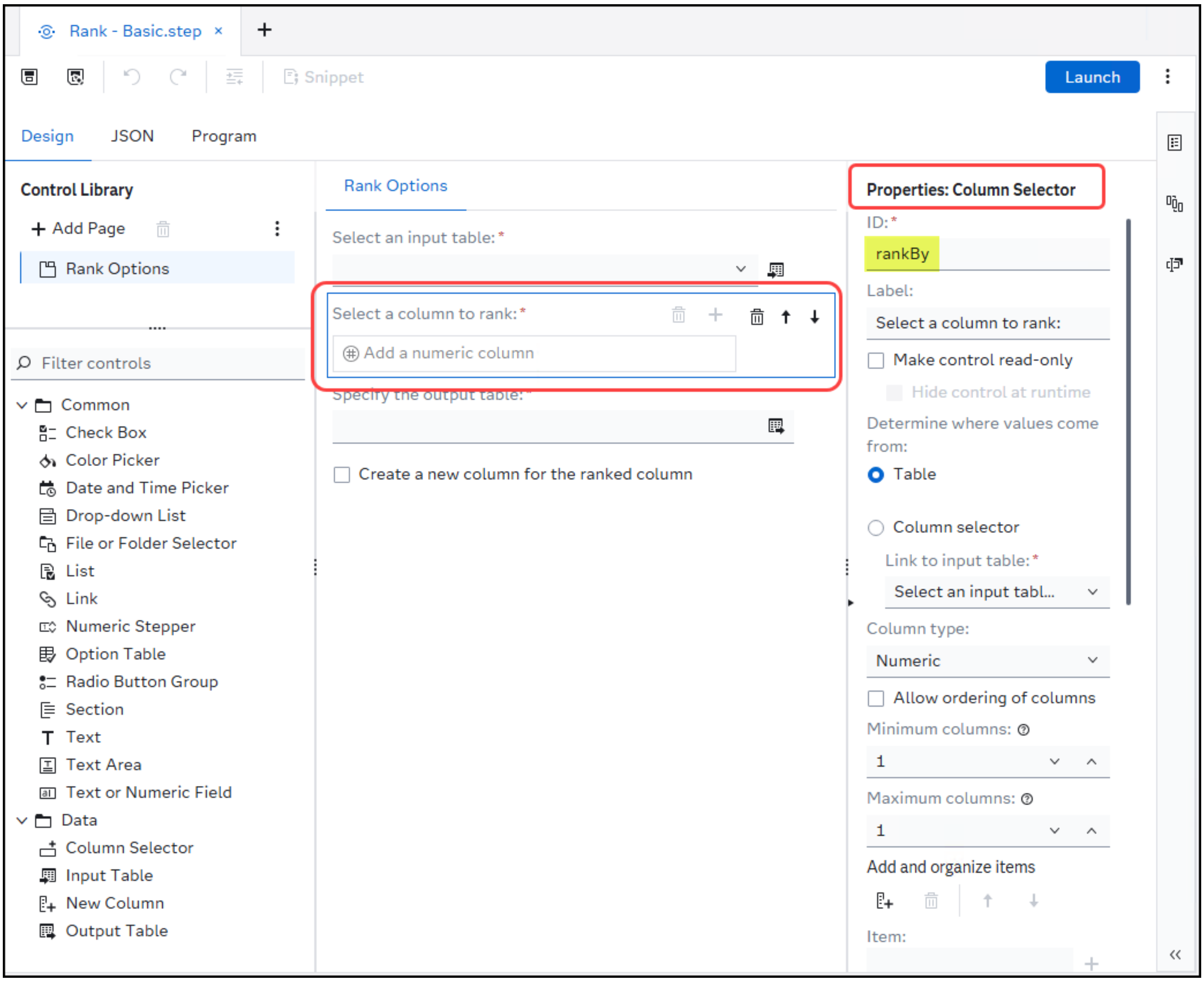
> ✎ There are also more complex *starter template* examples to review; however, for this exercise you will focus on the basic example.

2. Select the **Select an input table** control to view its properties.



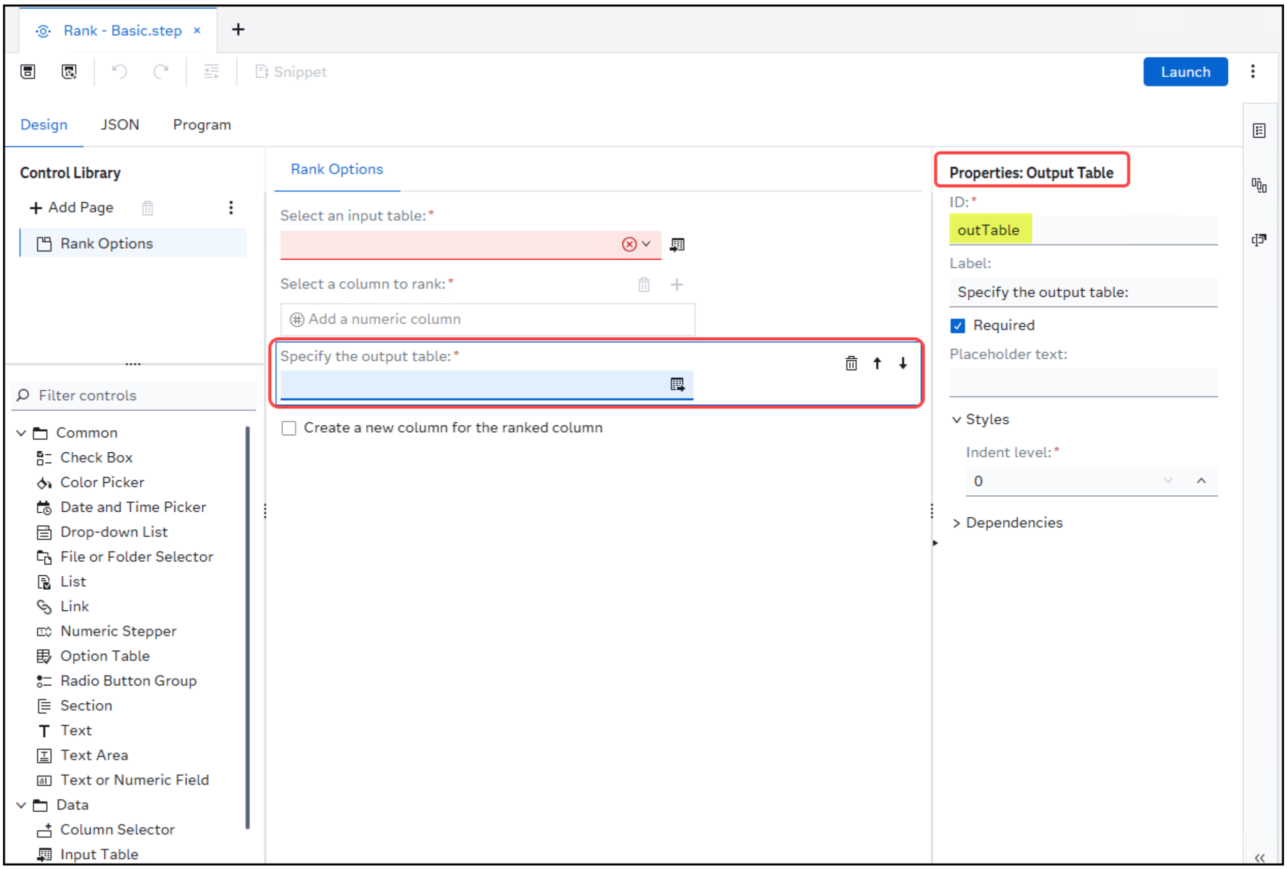
> ✎ The property *ID* for the control is **inTable**.

3. Select the **Add a numeric column** control to view its properties.



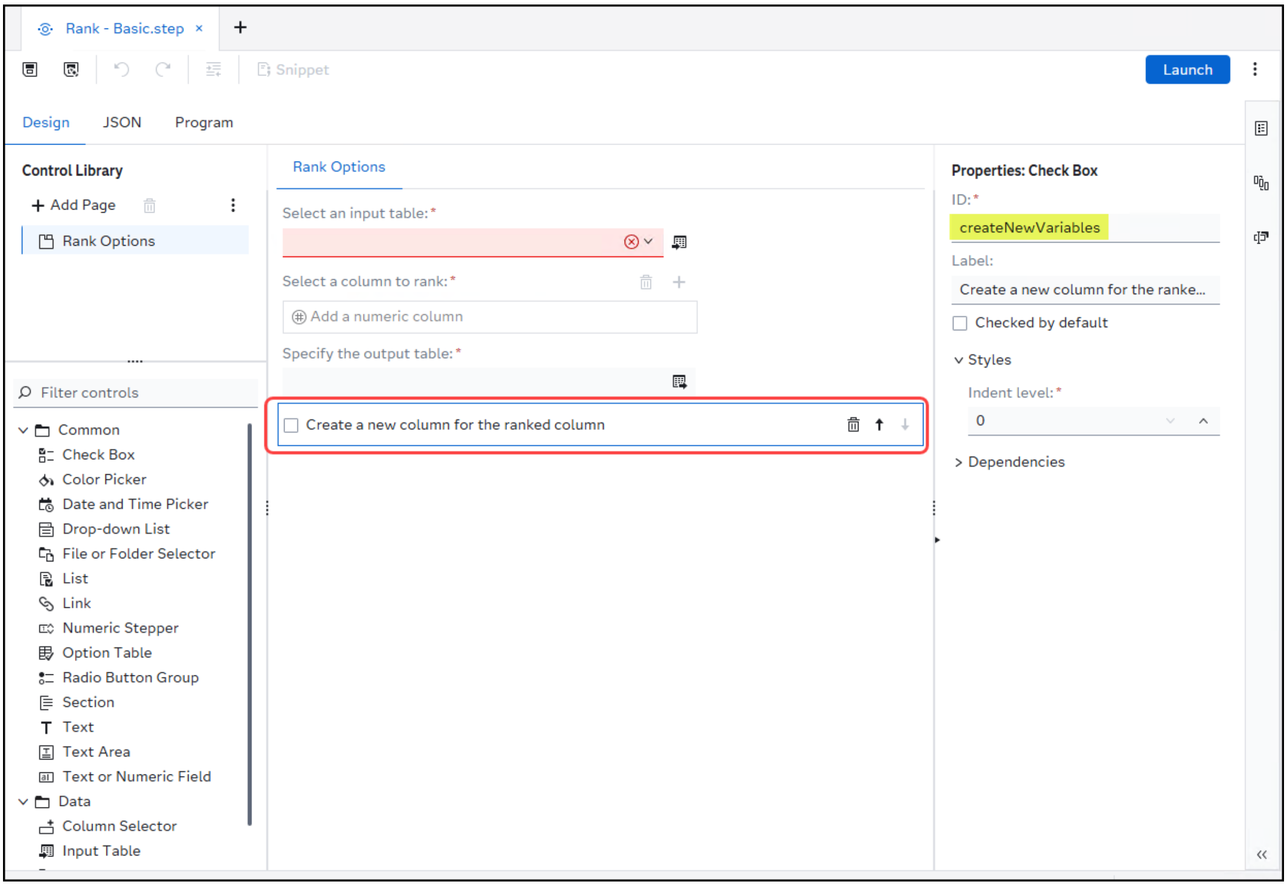
> ✎ The property *ID* for the control is **rankBy**.

4. Select the **Specify the output table** control to view its properties.



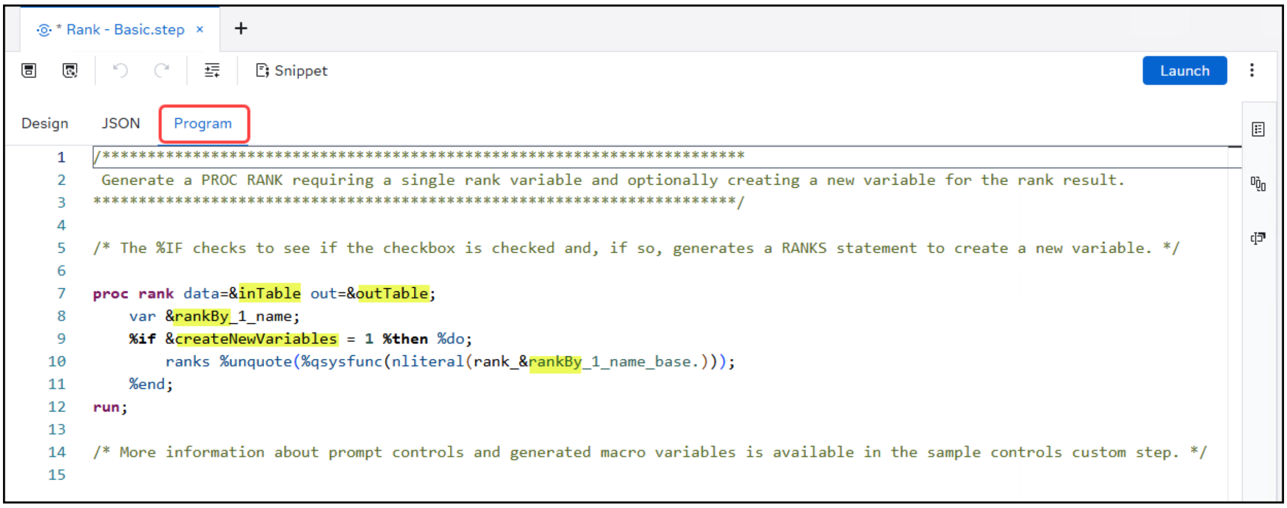
> ✎ The property *ID* for the control is **outTable**.

5. Select the **Create a new column for the ranked column** *check-box* control to view its properties.



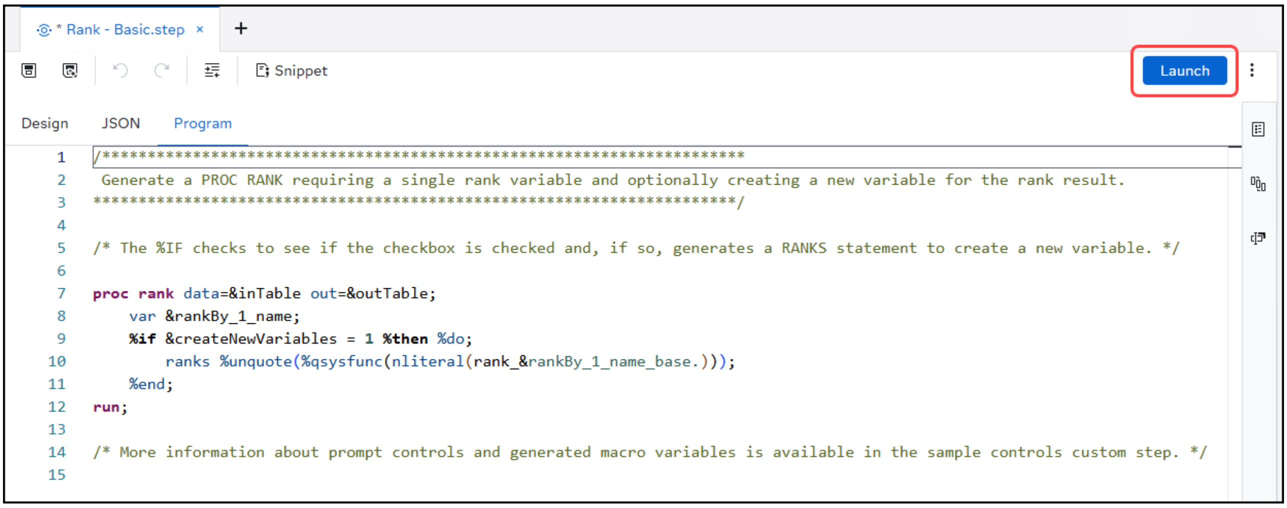
> ✎ The property *ID* for the control is **createNewVariables**.

6. Select the **Program** tab to view the code for the custom step.



> ✎ The property *ID* references for the various controls from the user interface design for the custom step.

7. Click **Launch** to test the sample custom step in stand-alone mode.



8. Make the following selections:

   - Select an input table: **SASHELP.BASEBALL**.

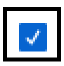
     > ✎ Click  to navigate to the desired input table.

   - Select a column to rank: **CrHits**

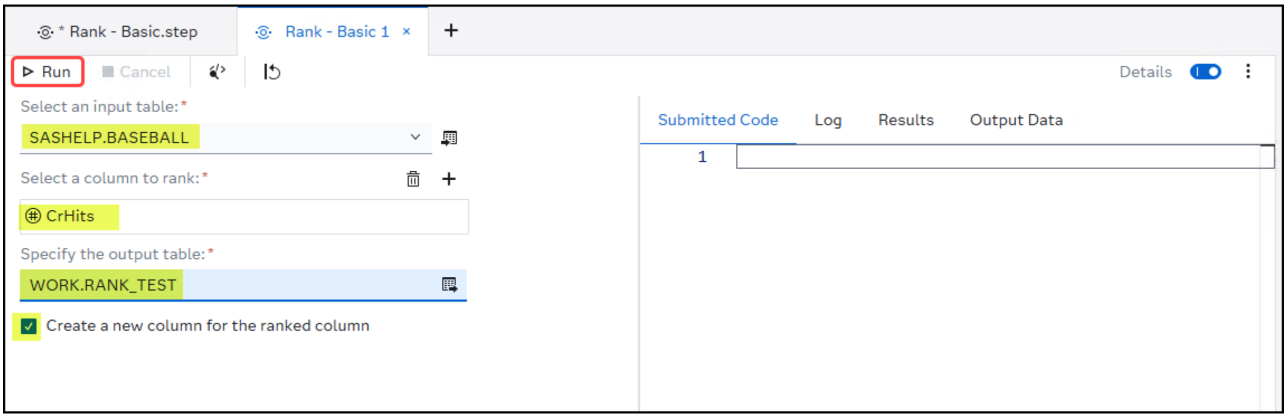
     > ✎ Click  to select a numeric column from the input table.

   - Specify an output table: **WORK.RANK_TEST**

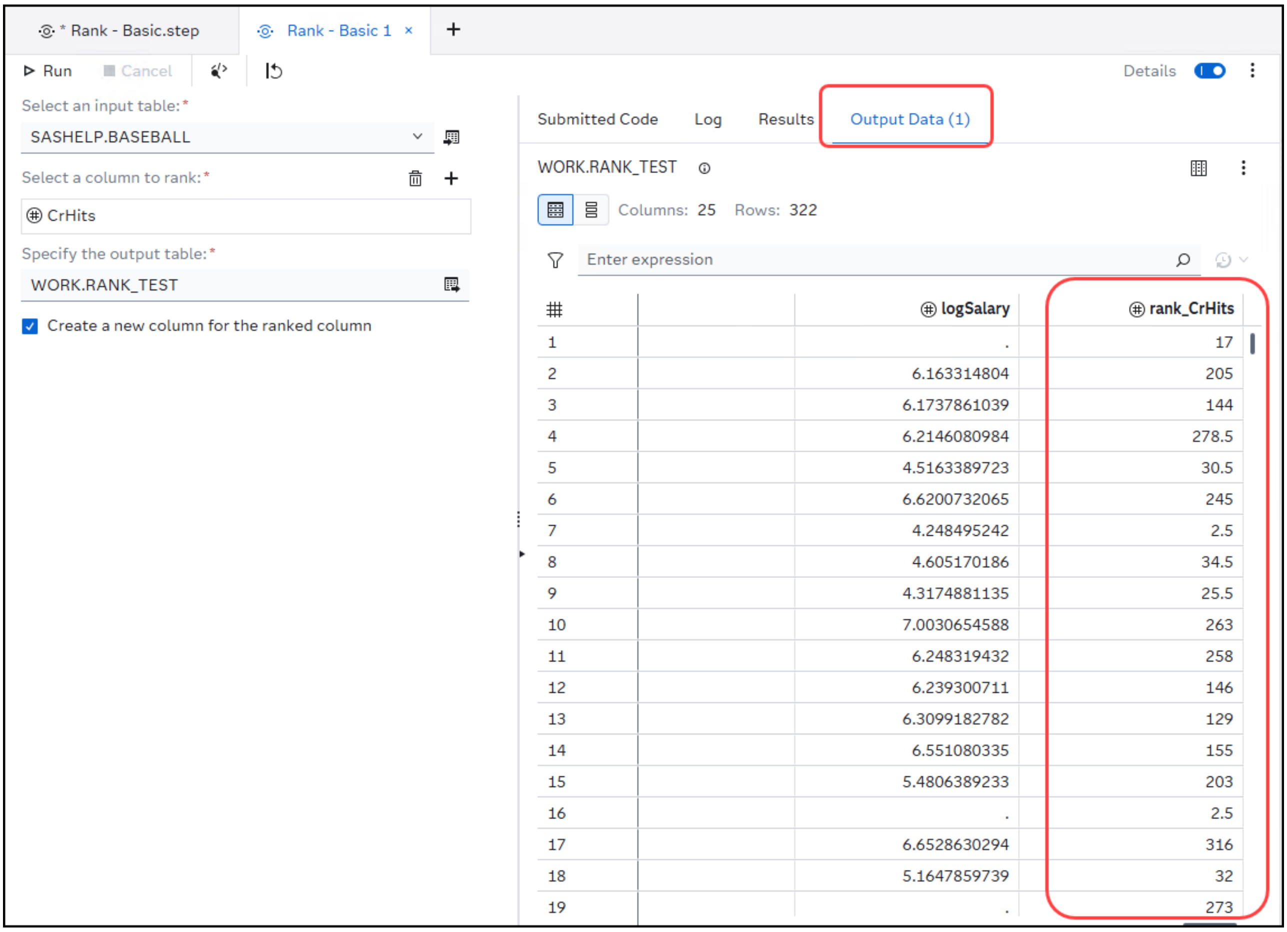
     > ✎ Click  to navigate to the desired output library and then enter the desired output table name.

   -  Create a new column for the ranked column

9. Click  to run the custom step.



10. Select the **Output Data (1)** tab to view the resulting output from the custom step.



11. Click **x** to close the *Rank - Basic 1* tab.

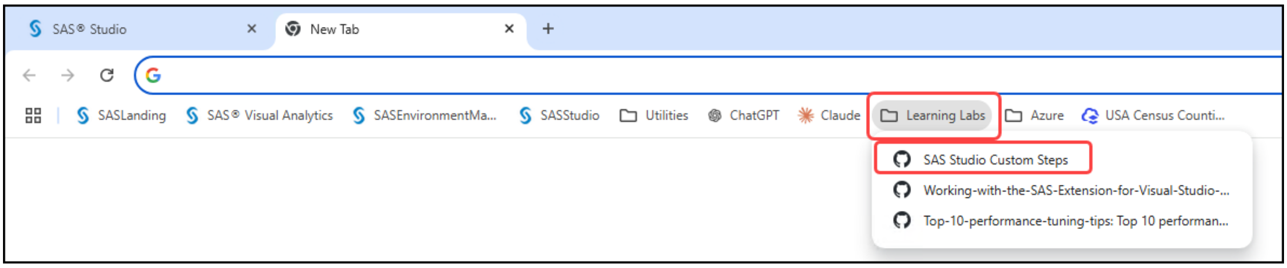
12. Click **x** to close the *Rank-Basic.step* tab.

13. Select **Don't Save** if prompted when closing the tab.

3. Search the Custom Step GitHub Repository

1. Open a new tab in the *Google Chrome* browser.

2. Select **Learning Labs ➔ SAS Studio Custom Steps** to open the SAS GitHub repository for custom steps.

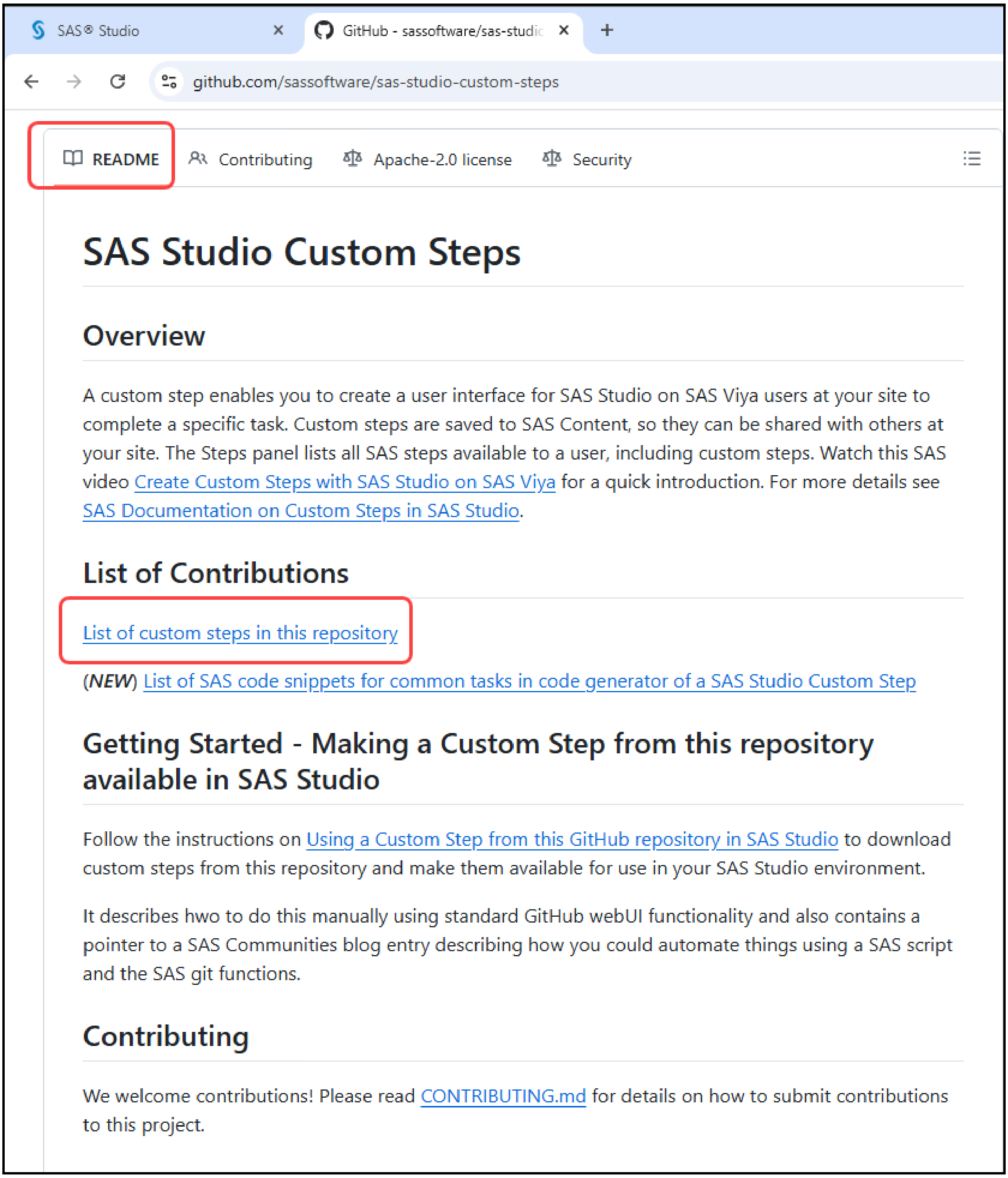


> ✏️ This is a saved link to the SAS Studio Custom Steps GitHub repository: https://github.com/sassoftware/sas-studio-custom-steps.
> ✏️ This is a SAS Community article on how you can download all the custom steps for this repository: https://communities.sas.com/t5/SAS-Communities-Library/Uploading-all-Custom-Steps-from-the-SAS-Software-GitHub/ta-p/887903

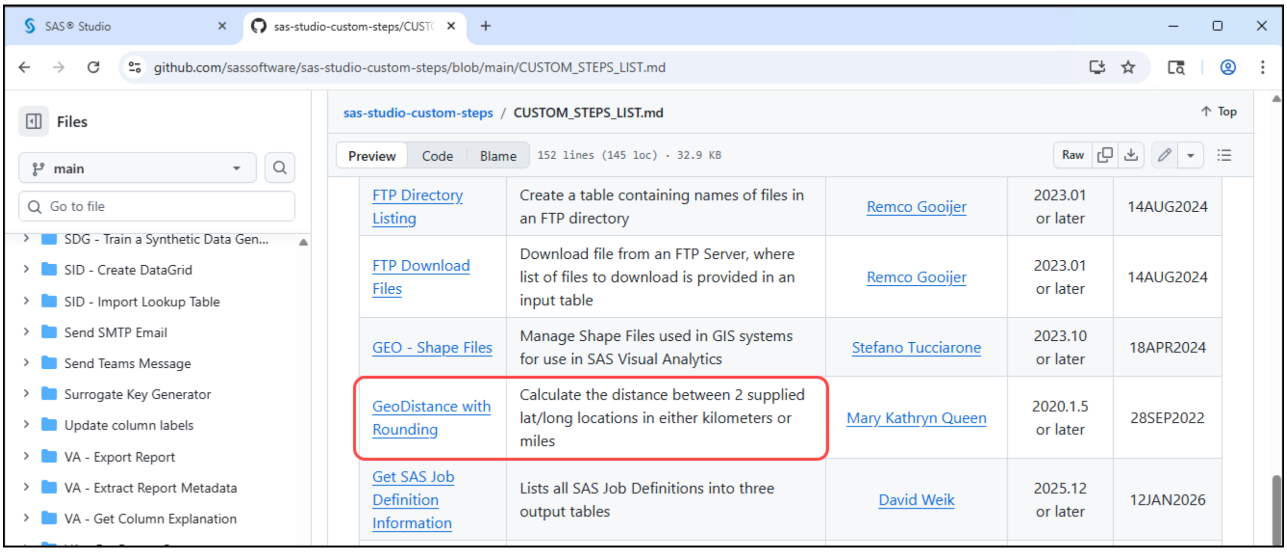
3. Click **Sign In** if prompted to sign in to the repository.

4. Scroll down towards to the bottom of the page to view the **README** information for the repository.

5. Select the link for **List of custom steps in this repository**.

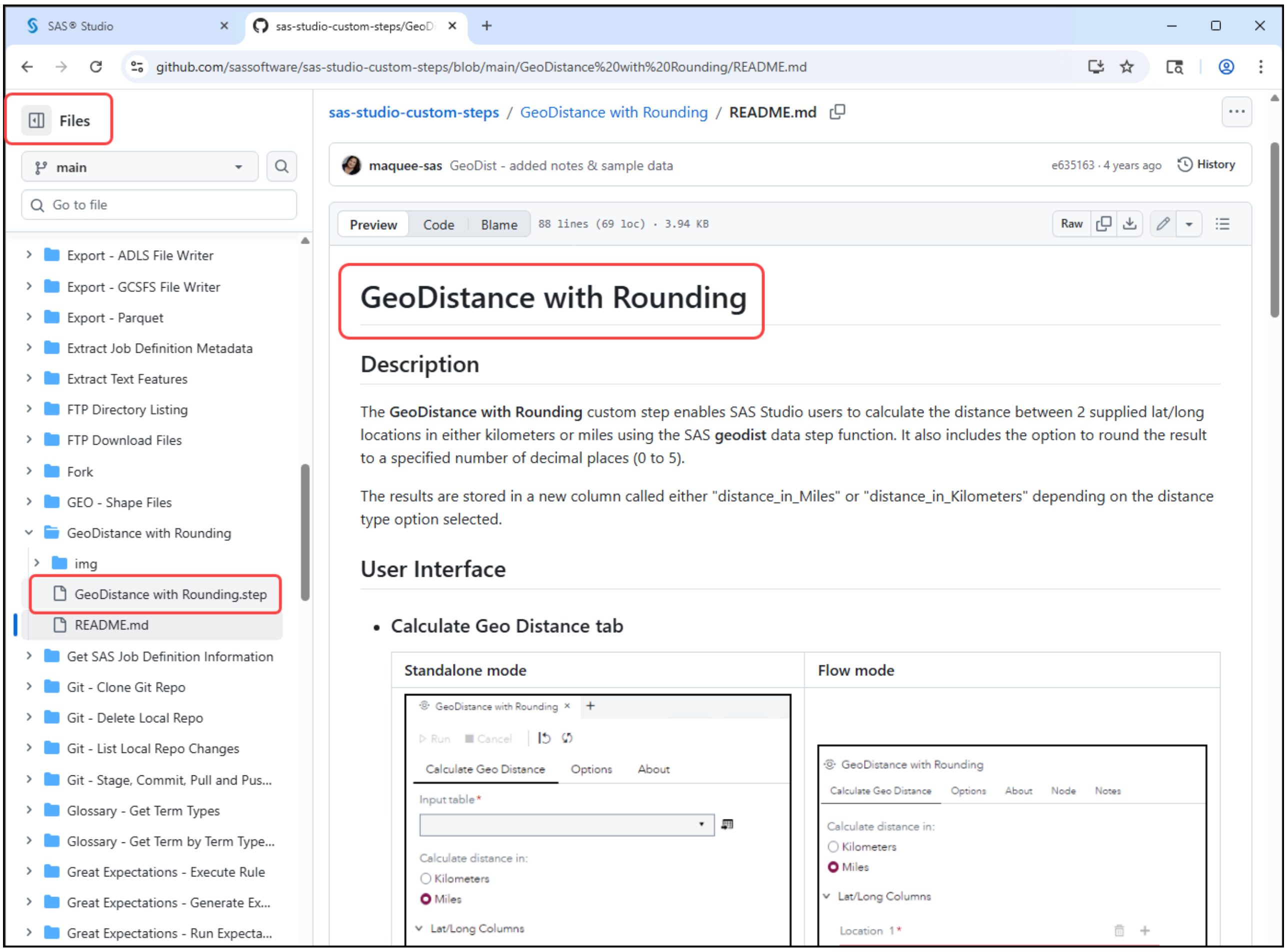


6. Scroll down until you locate the entry for **GeoDistance with Rounding** in the table of *Available Custom Steps*.

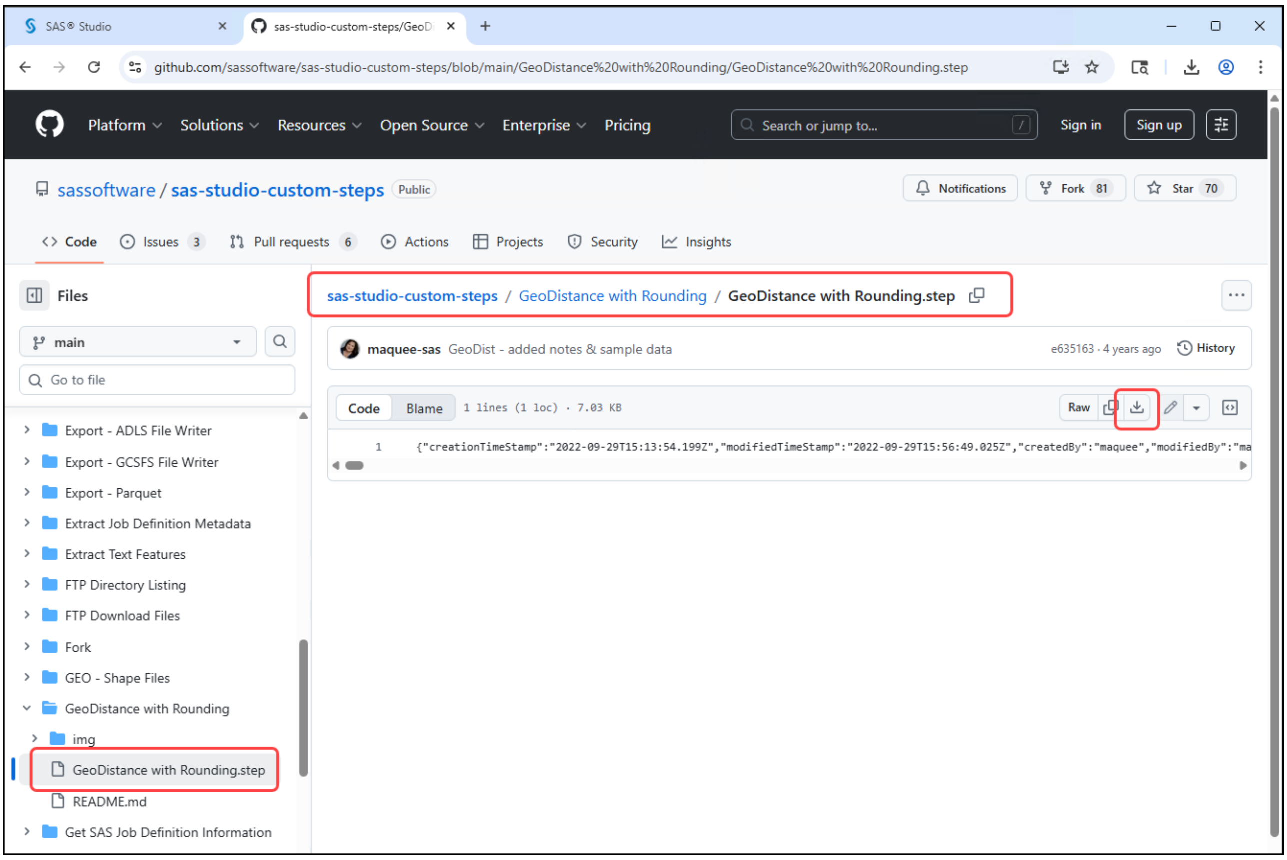


7. Select the link for **GeoDistance with Rounding** to view additional descriptive information about the custom step.

8. On the left-hand side in the *Files* section, select the **GeoDistance with Rounding.step**.



9. Click  to download the file for this custom step.

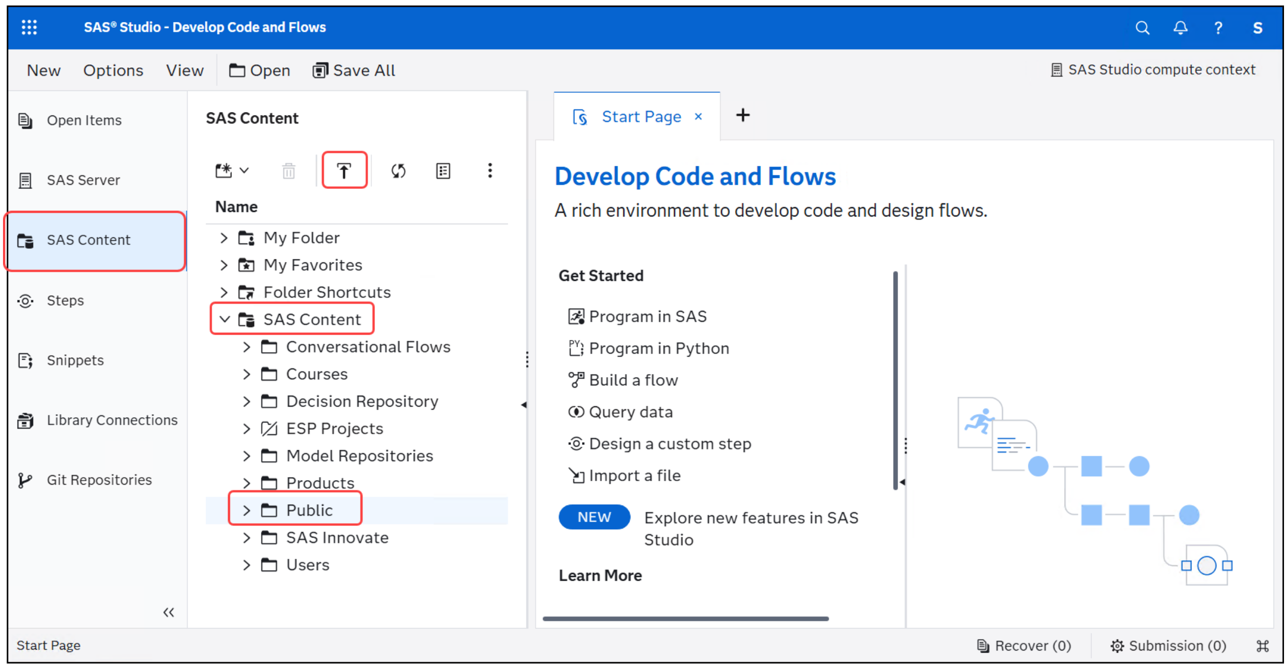


10. Return to the **SAS Studio** tab in the *Google Chrome* browser.

11. Select  to view the *SAS Content* folders.

12. Expand the *SAS Content* folder and select the **Public** folder.

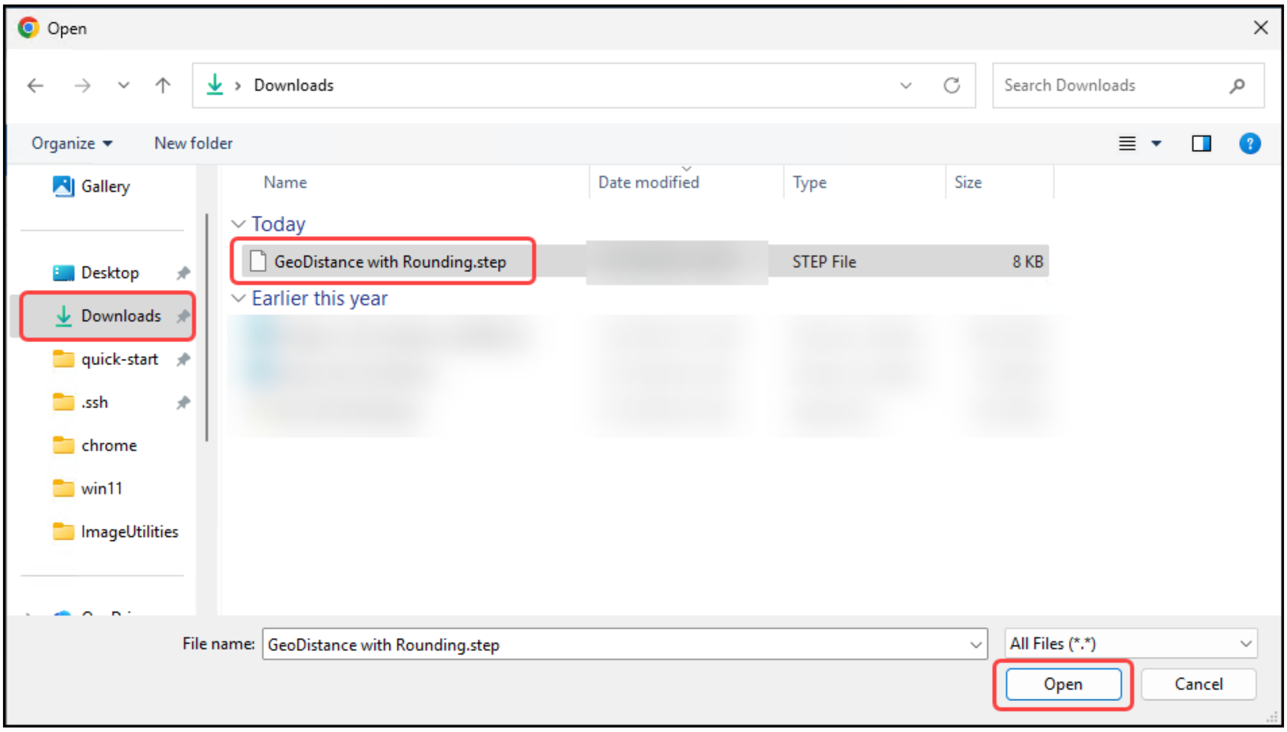
13. Click  to upload files to the selected folder.



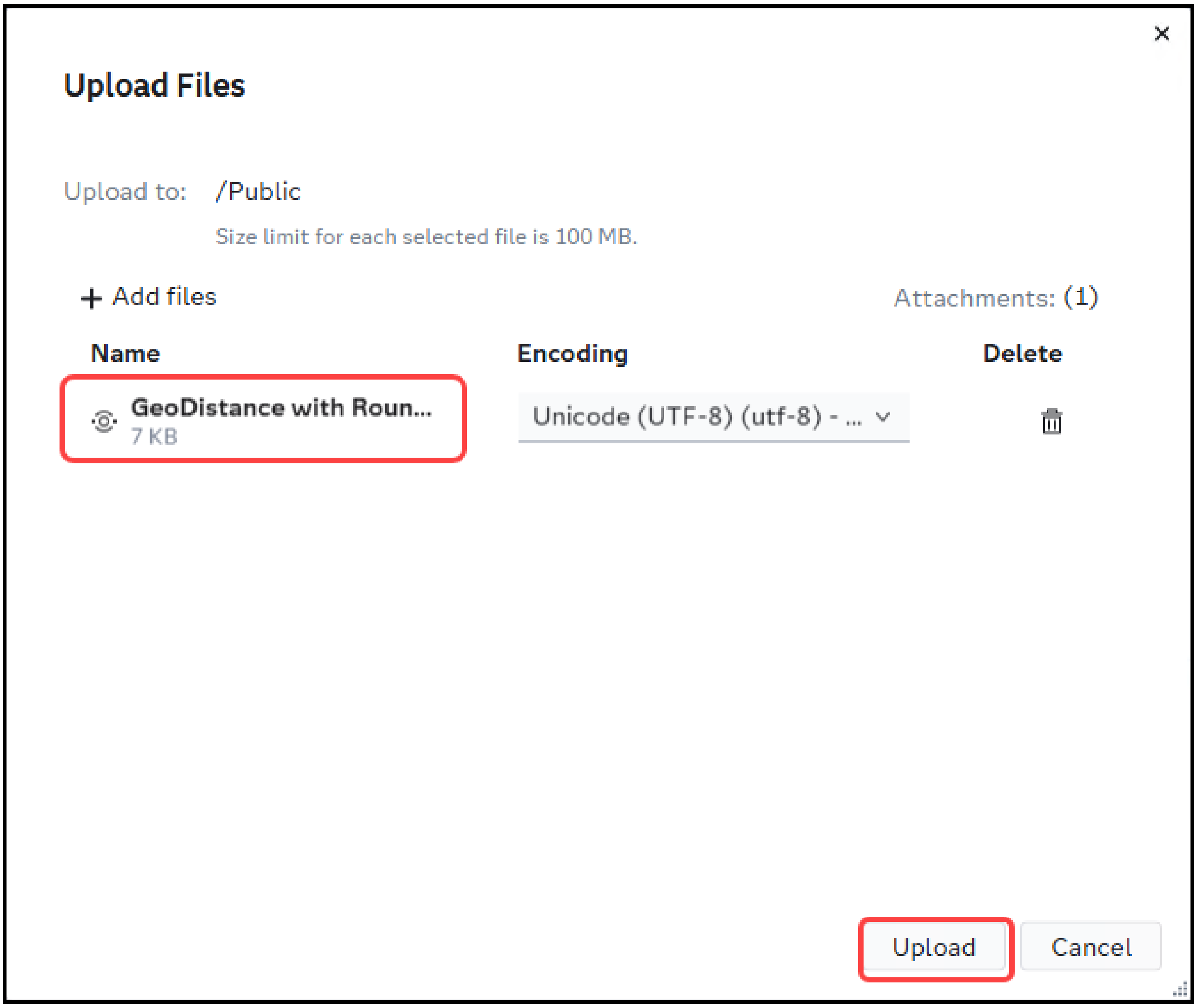
14. Click 

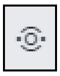
and navigate to **Downloads** folder.

15. Select the **GeoDistance with Rounding.step** file and click **Open**.

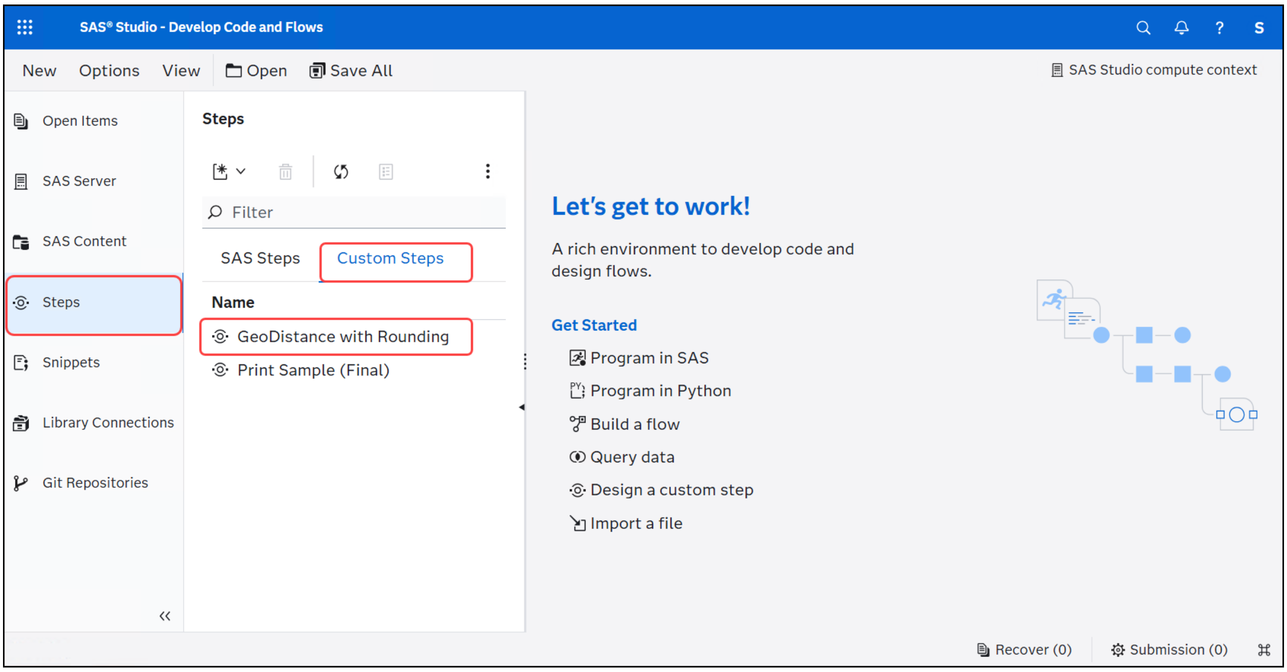


16. Click **Upload** to upload the attached file(s).



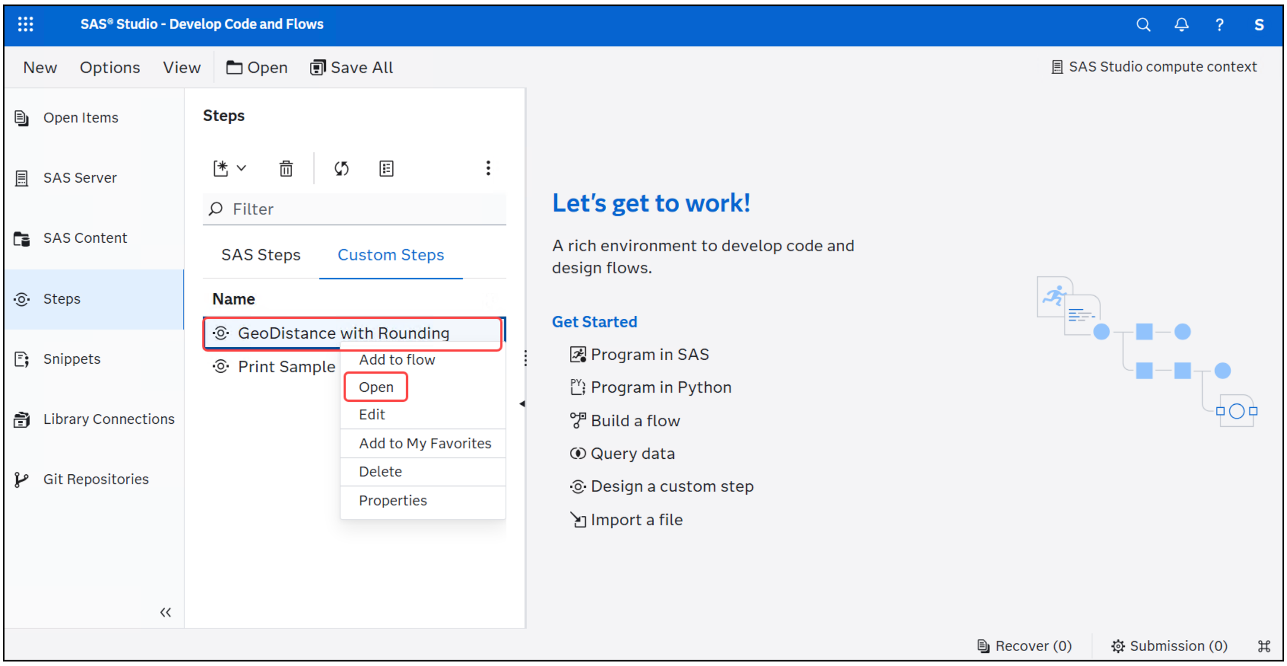
17. Select  to view the **Steps** pane.

18. Select the **Custom Steps** to confirm the *GeoDistance with Rounding* custom step is listed.



4. Test the Custom Step using Stand-alone Mode

1. In the *Steps* pane, select the **GeoDistance with Rounding** custom step and right-click then select **Open** to open the custom step in stand-alone mode.

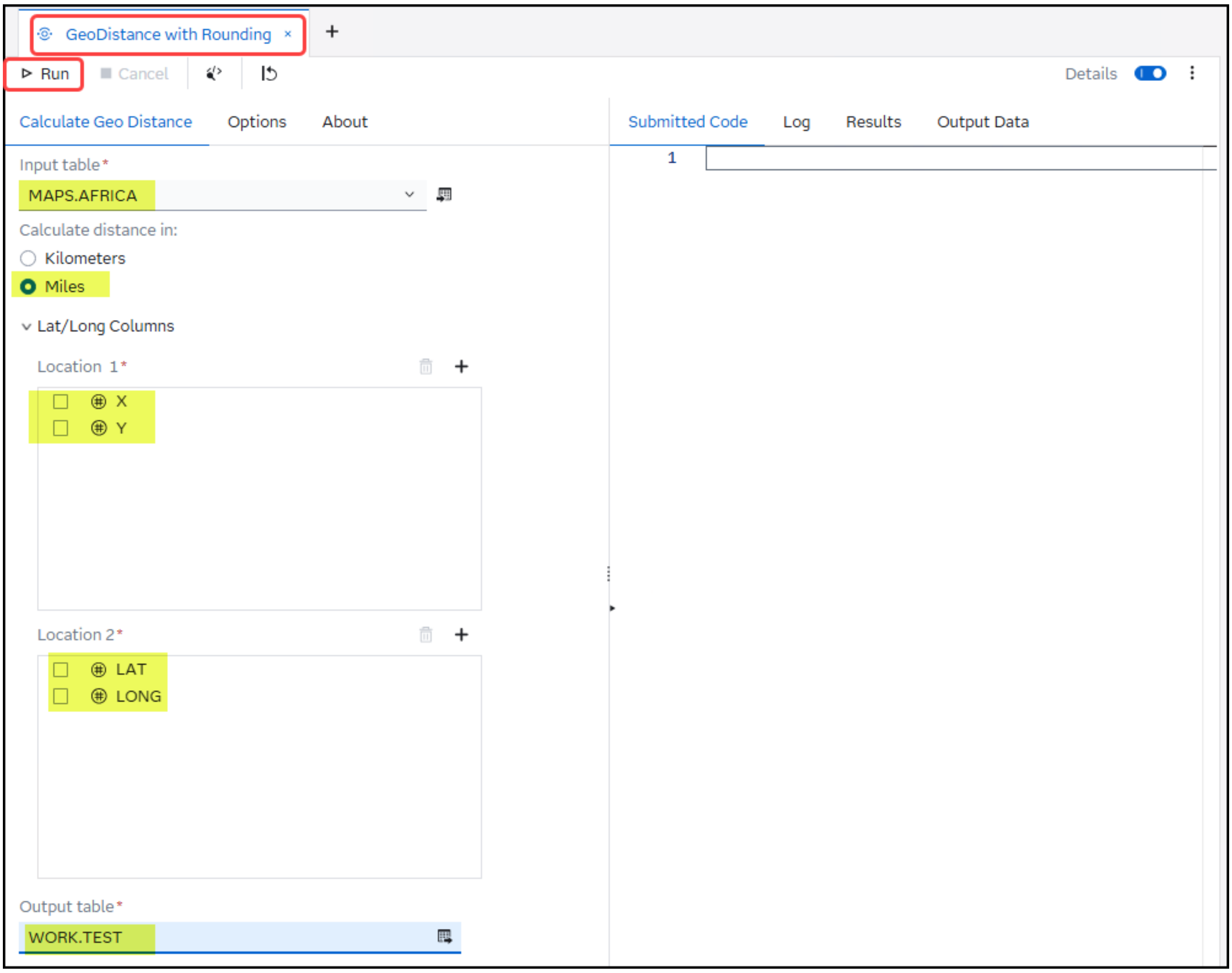


2. Select the following required information for the custom step:

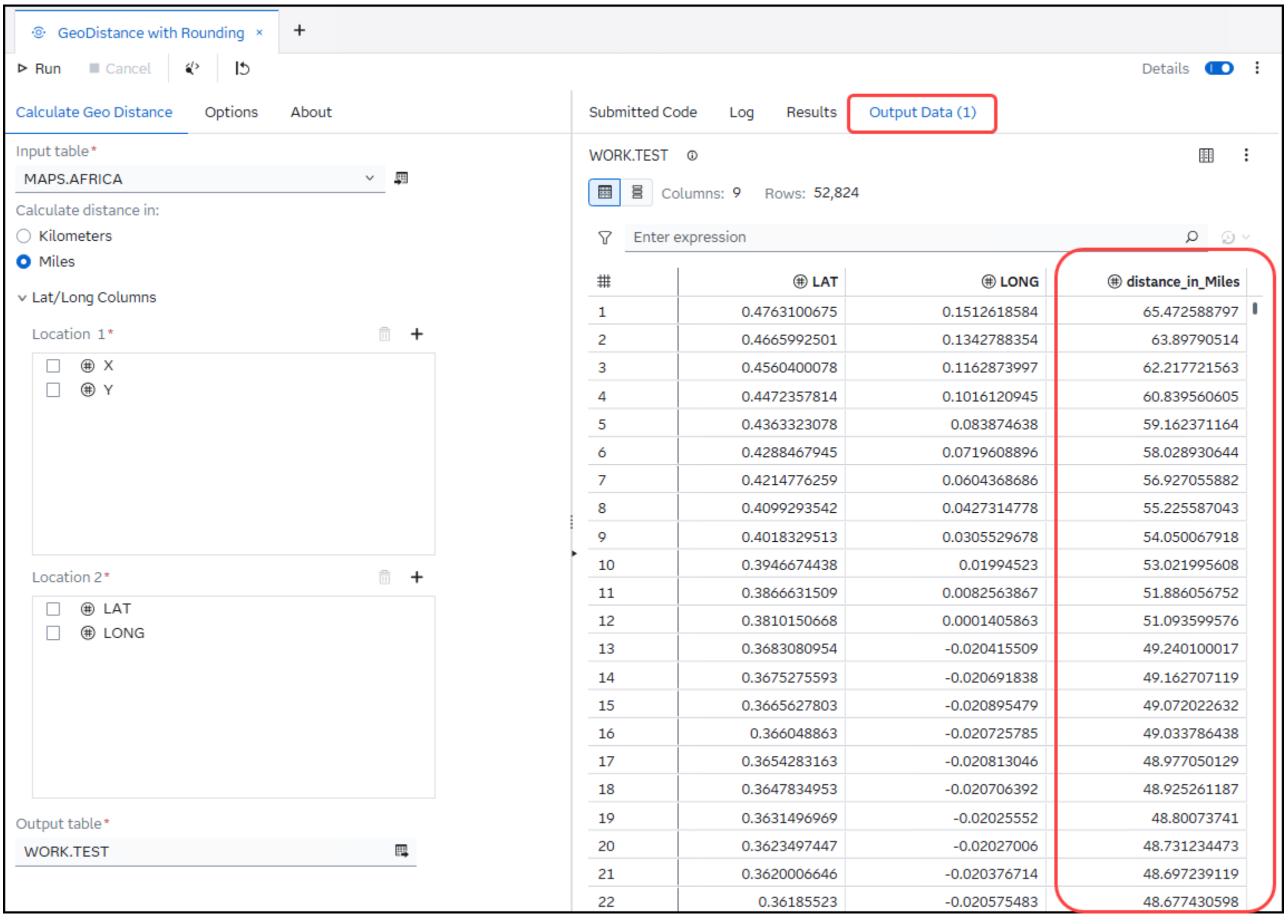
   - Input table: **MAPS.AFRICA**
   - Calculate distance in: **Miles**
   - Location 1:

        - **X**
        - **Y**
    - Location 2:
        - **LAT**
        - **LONG**
    - Output table: **WORK.TEST**

3. Click ▷ Run to run the custom step in stand-alone mode.



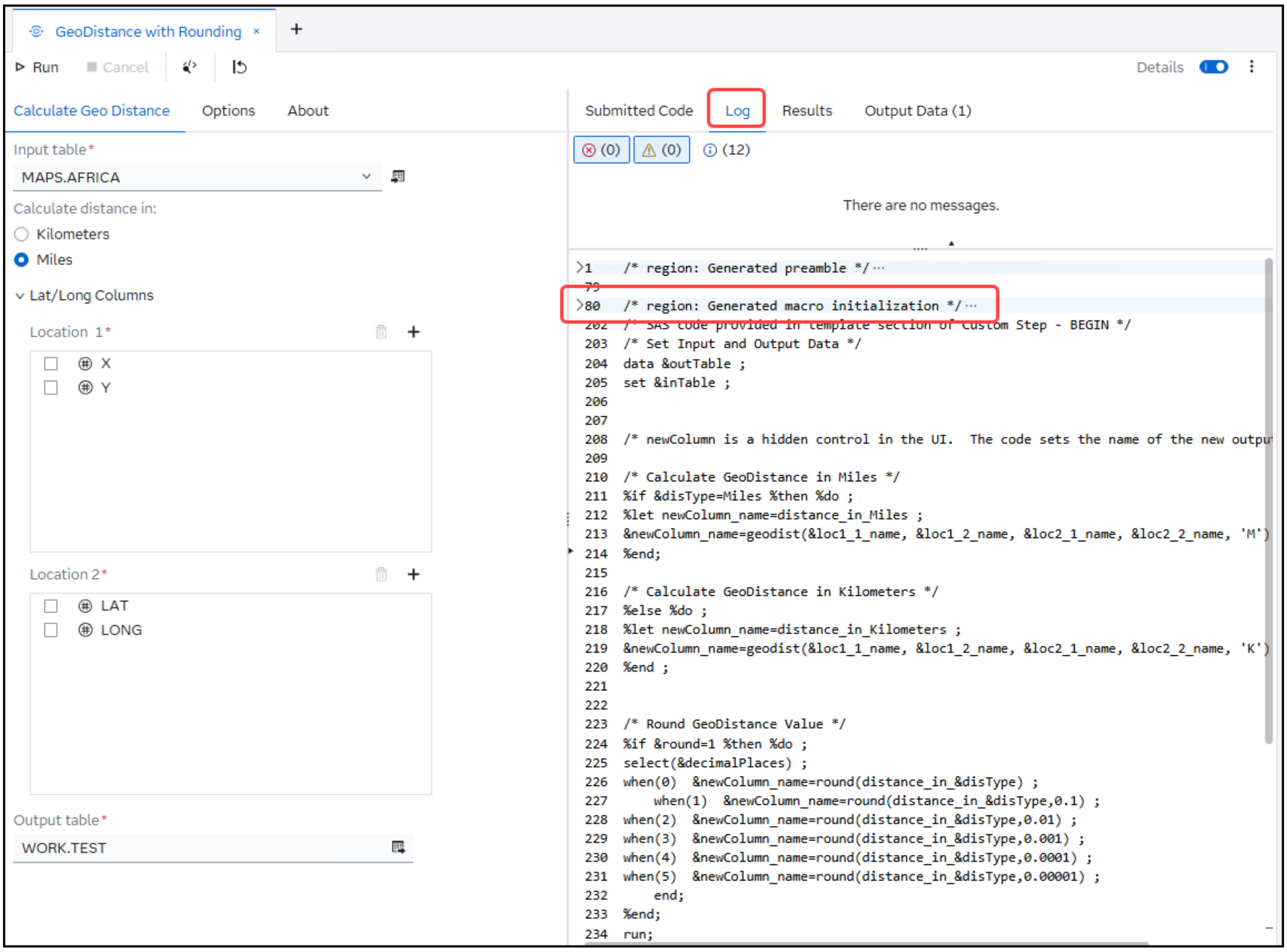
4. Review the **Output Data** tab.



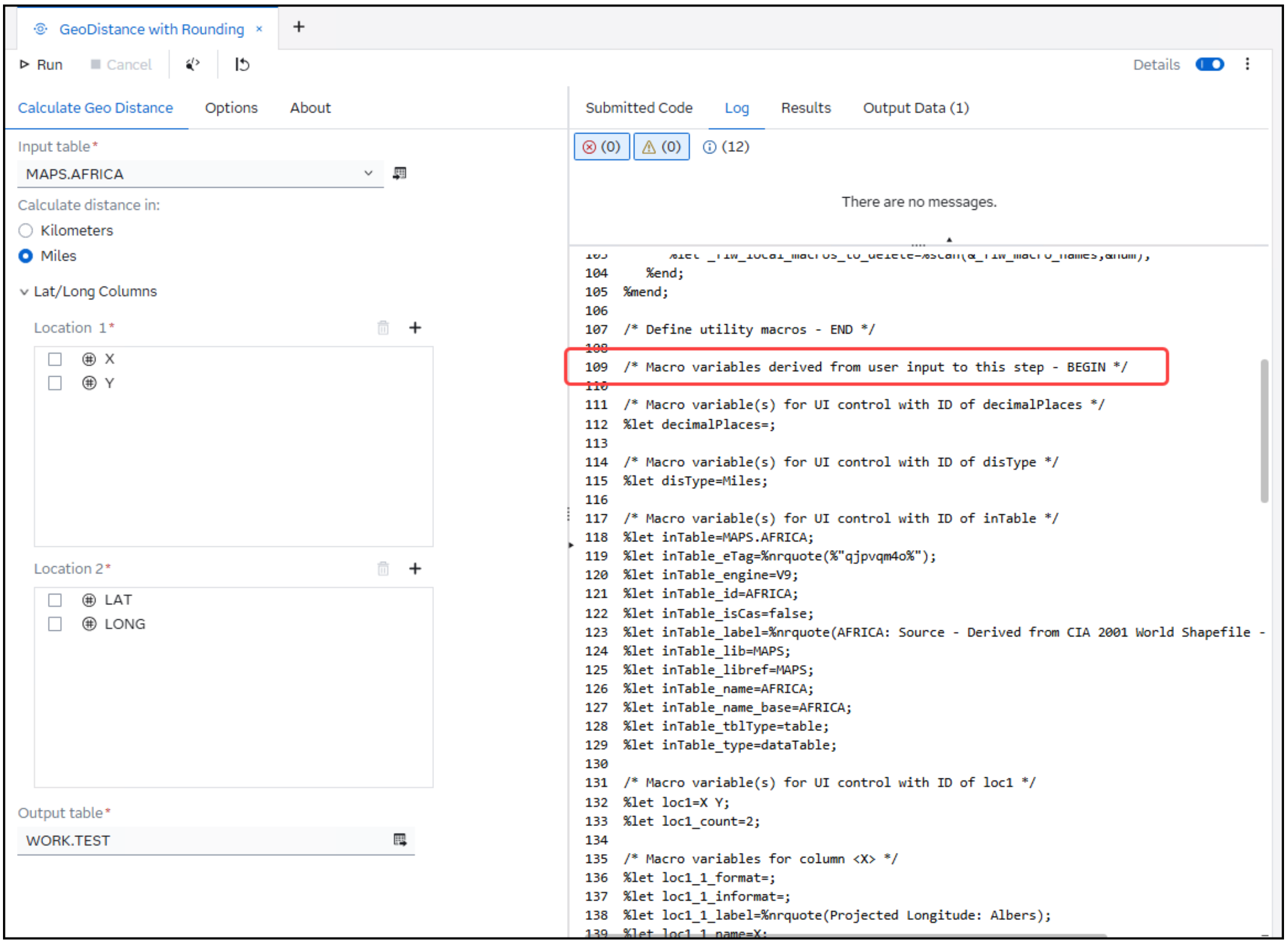
5. Review the Log for Macro Variable Names and Values

    1. Select the **Log** tab.

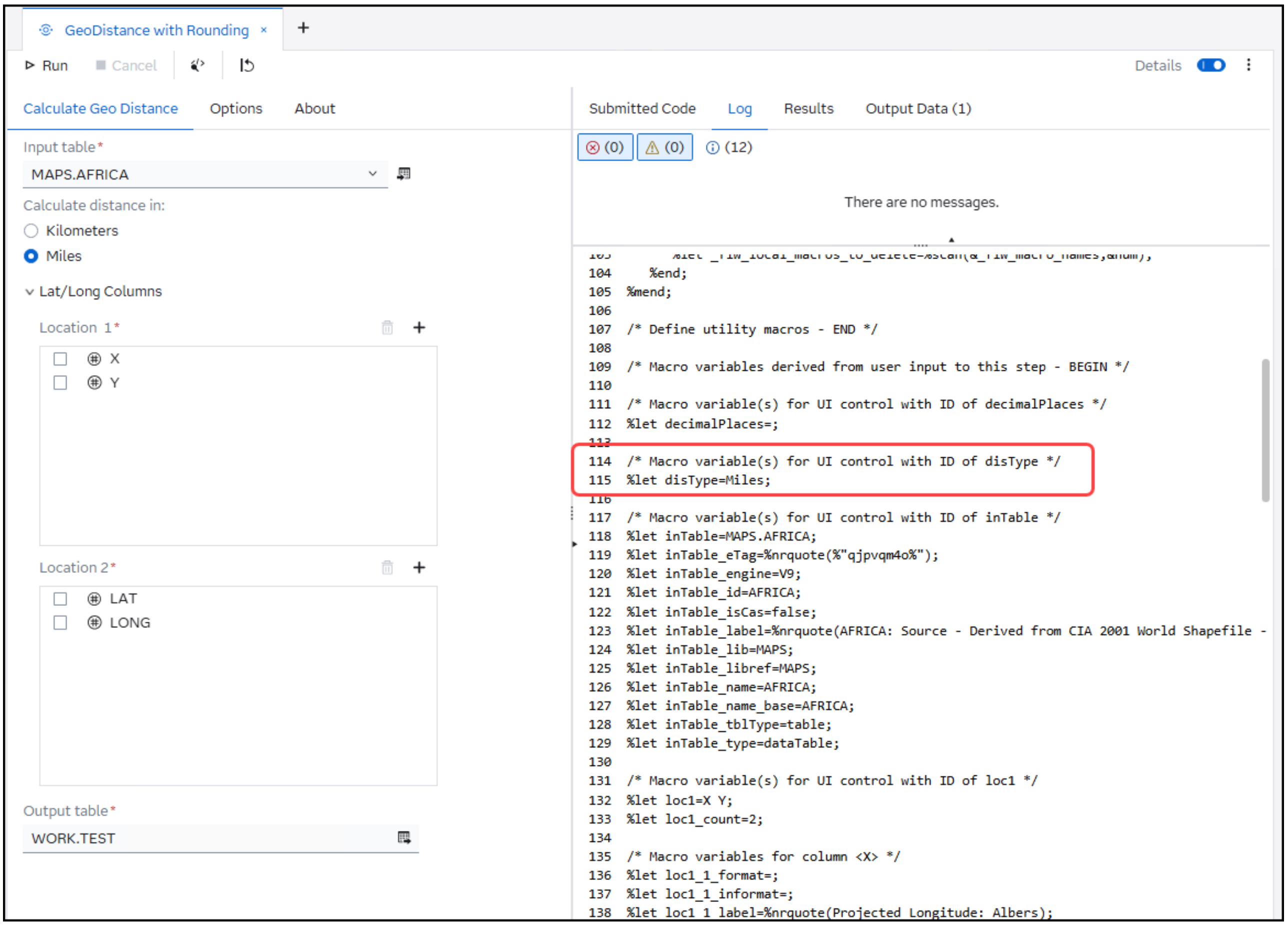
    2. Expand the section named **/*region: Generated macro initialization*/**.



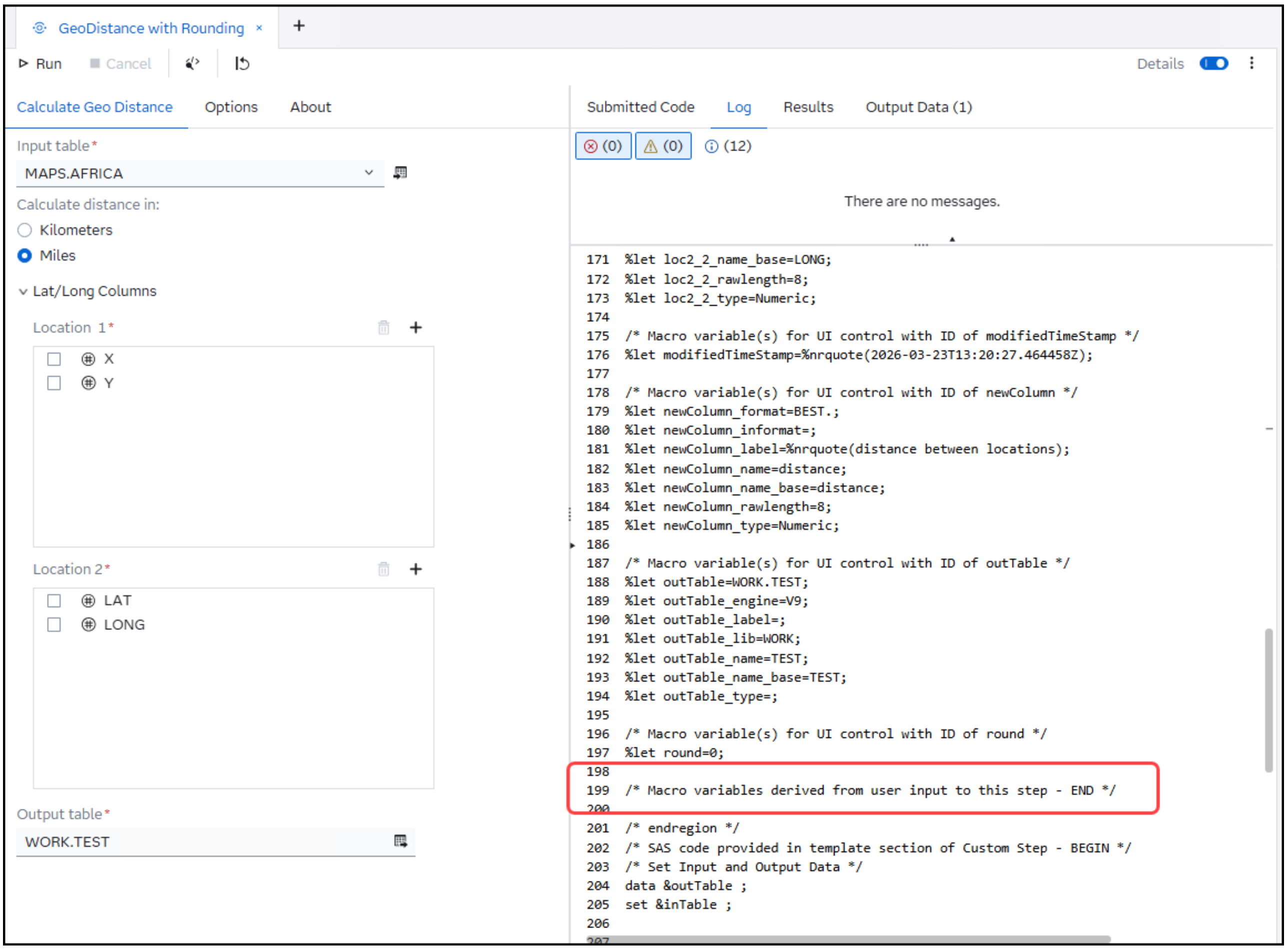
3. Scroll up to the section heading named **/* Macro variables derived from user input to this step - BEGIN */** to view macro variable names and current values for the user input from the custom step.



4. Notice that the macro variable **disType** is set to **Miles**.



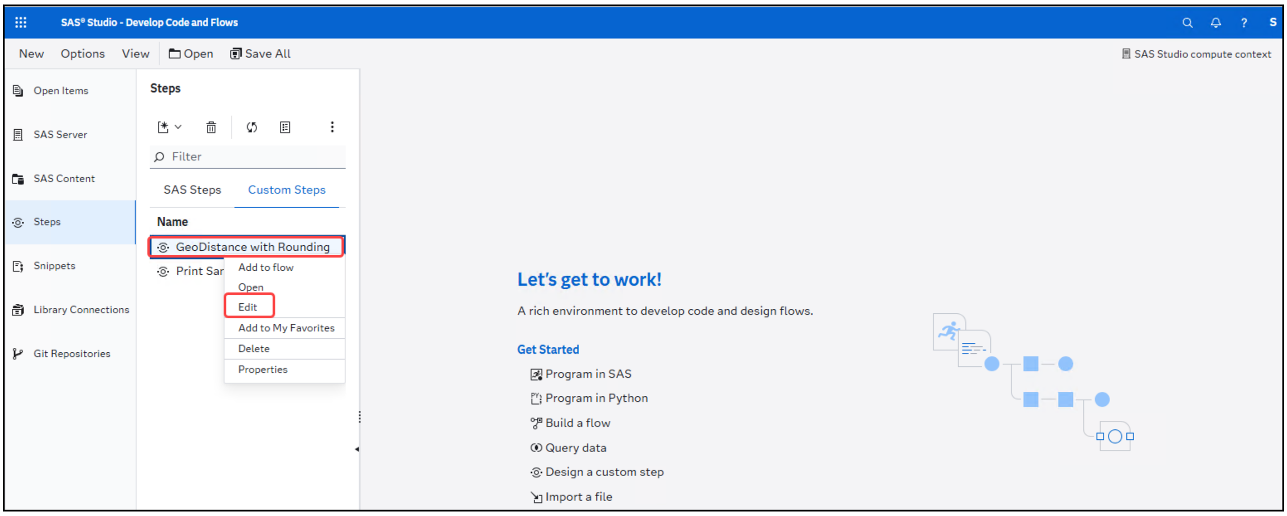
5. Scroll to the section closer named **/* Macro variables derived from user input to this step - END */**. This ends the section of the macro variable names and values for the user input from the custom step.



6. Click **x** to close the open custom step in stand-alone mode.

6. Use **%put** Statements to Aid in the Testing of your Code

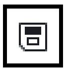
1. In the *Steps* pane, select the **GeoDistance with Rounding** custom step and right-click then select **Edit** to open the custom step in edit mode.



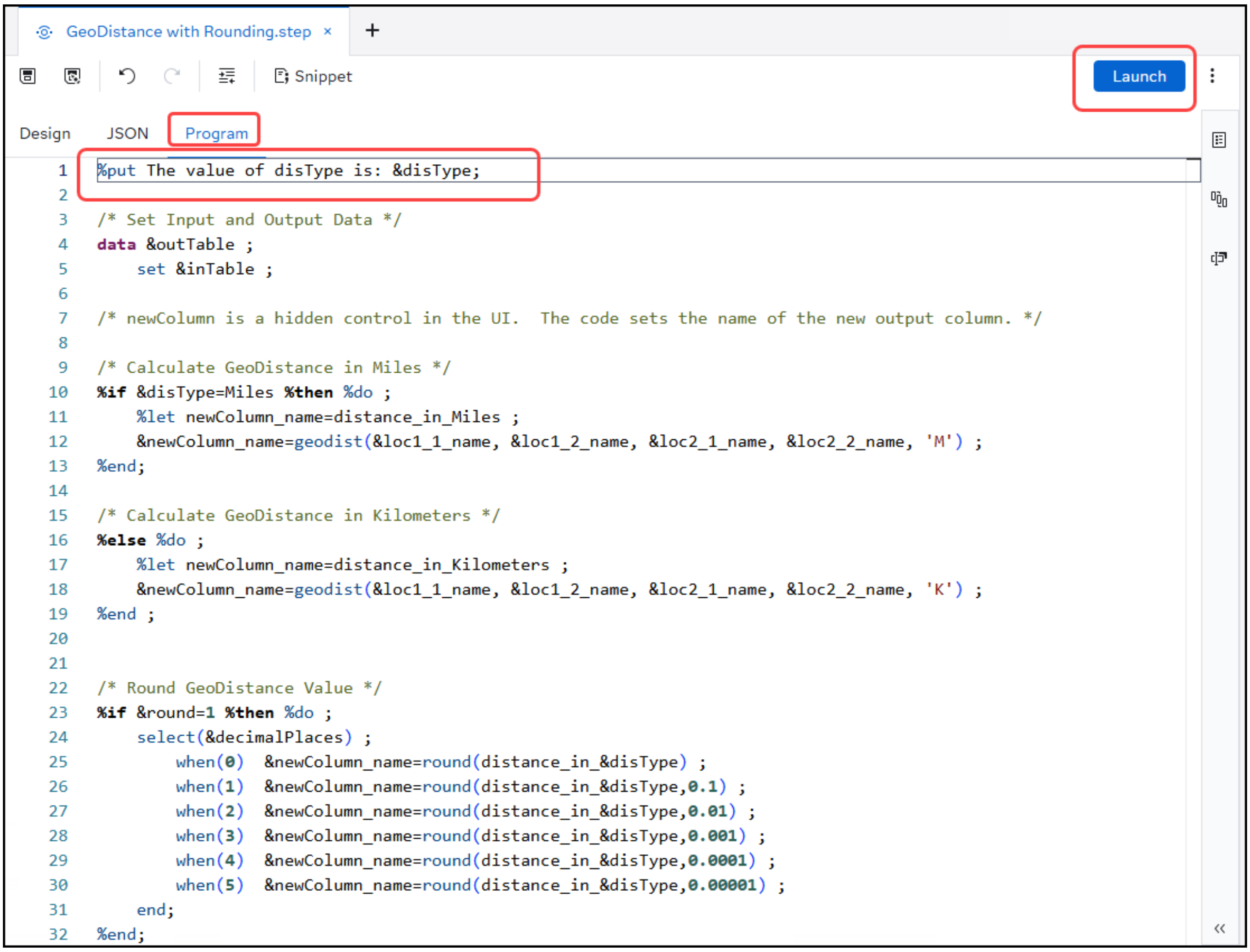
2. Select the **Program** tab to view the code for the custom step.

3. Add the following line at the beginning of the code block:

```
%put The value of disType is: &disType;
```

4. Click  to save the changes to the program code.

5. Click **Launch** to test the custom step.



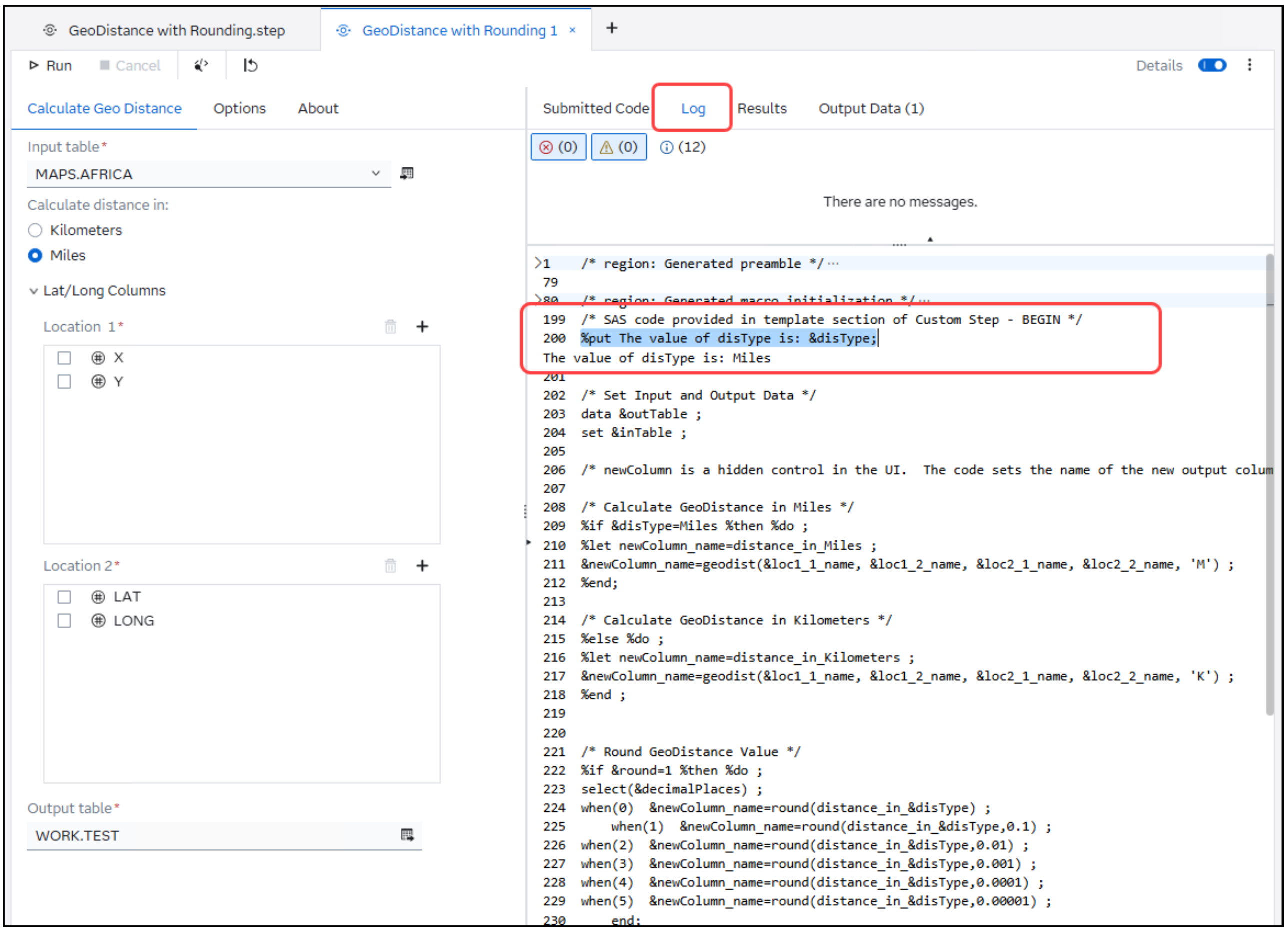
6. Select the following required information to test the custom step:

    - Input table: **MAPS.AFRICA**
    - Calculate distance in: **Miles**
    - Location 1:
        - **X**
        - **Y**
    - Location 2:
        - **LAT**
        - **LONG**
    - Output table: **WORK.TEST**

7. Click 

to run the custom step in stand-alone mode.

8. Review the **Log** tab and scroll up to find the **%put The value of disType is: &disType;** statement.

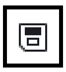


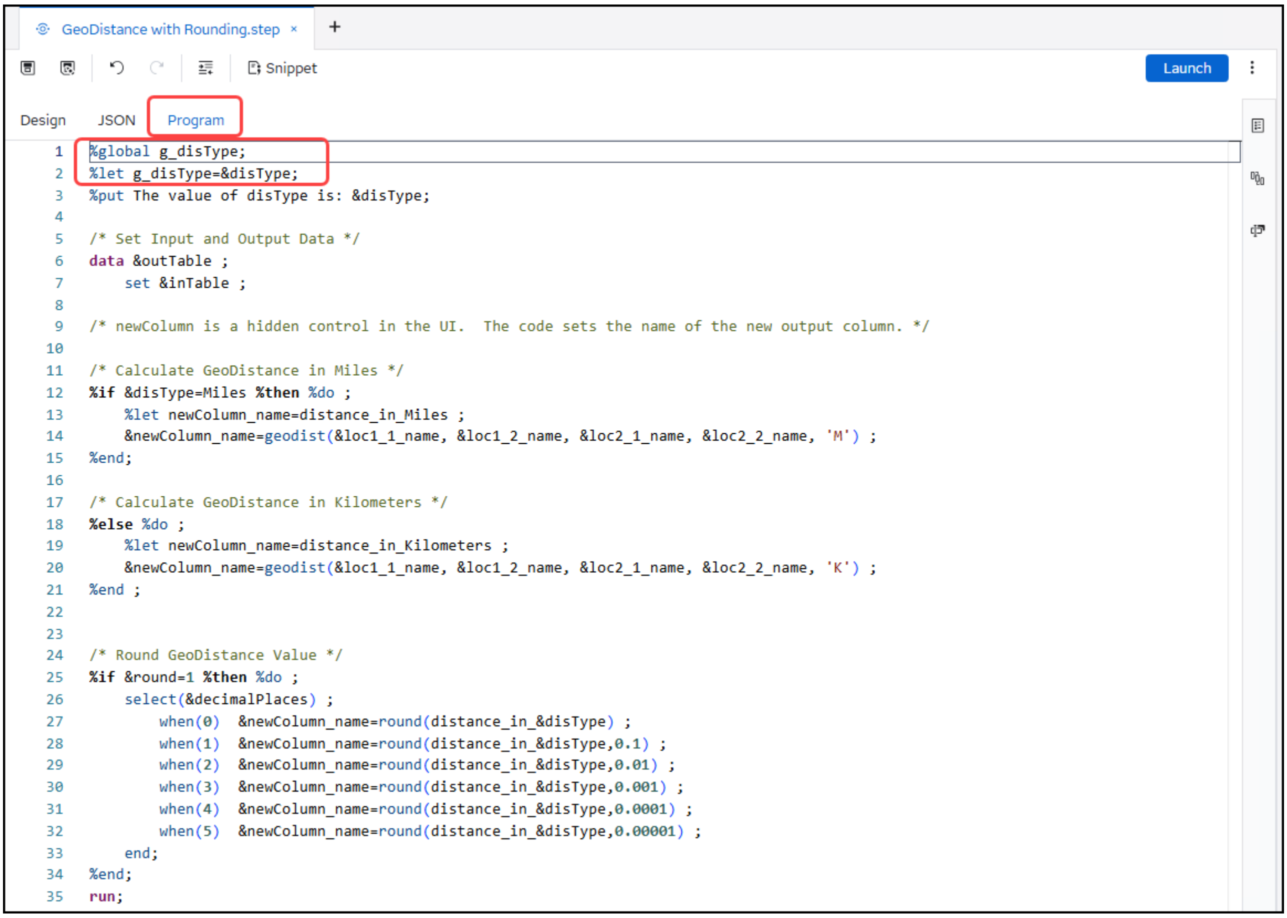
9. Click **x** to close the *GeoDistance with Rounding 1* tab.

7. Use **%global** Statement to Assign Macro Variable Values for Use Outside of the Custom Step

1. On the **Program** tab, add the following lines of code at the beginning of the code block:

```
%global g_disType;
%let g_disType=&disType;
```

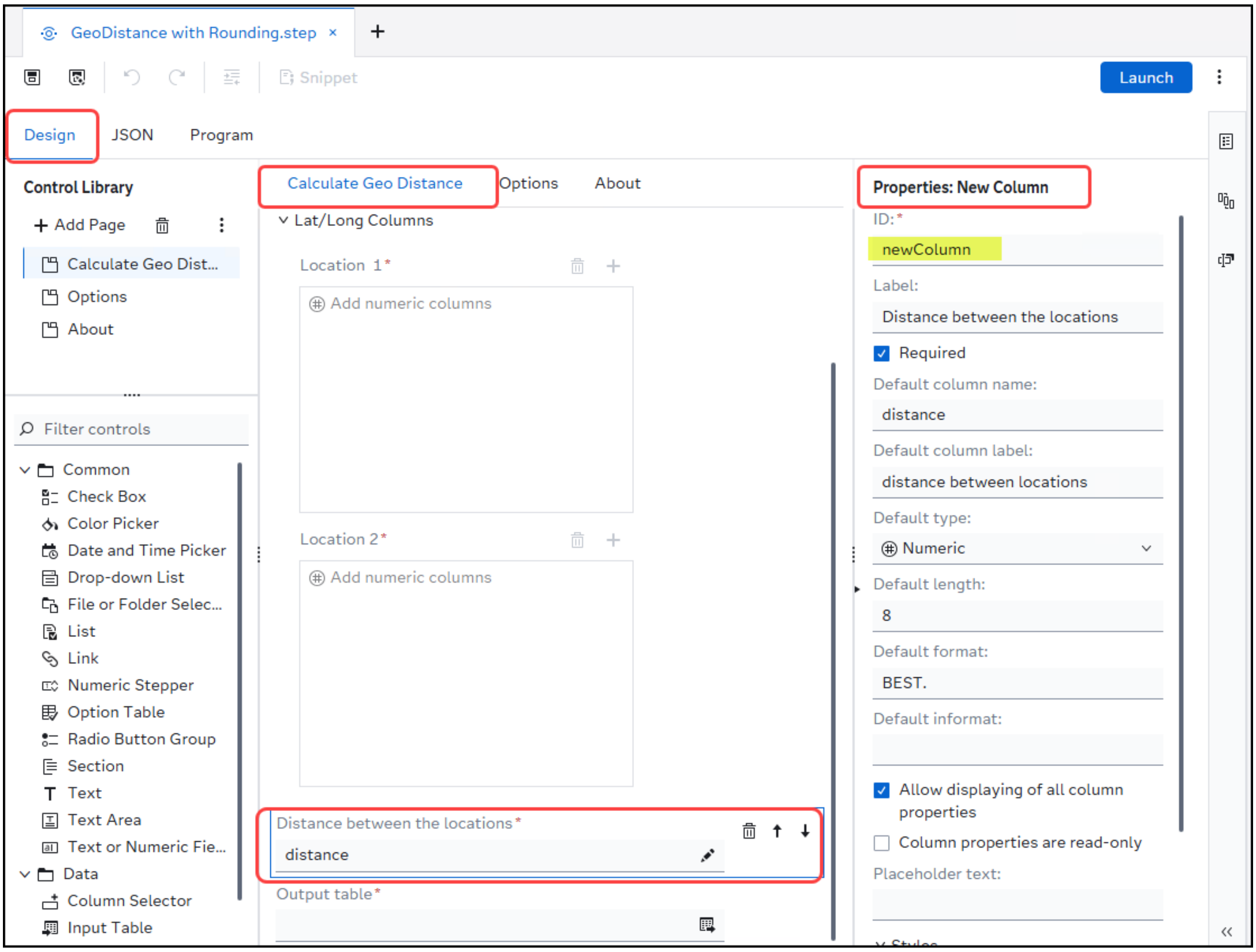
2. Click  to save the changes to the program code.



> ✎ Now, the global macro variable **&g_disType** can be referenced outside of the custom step in a SAS Studio flow.

8. Use the New Column Control (if your code creates a new column)

1. Select the **Design** tab to view the page design(s) for the custom step.
2. Select the **Calculate Geo Distance** page tab.
3. Select the **Distance between the locations** *new column* control to view its properties.

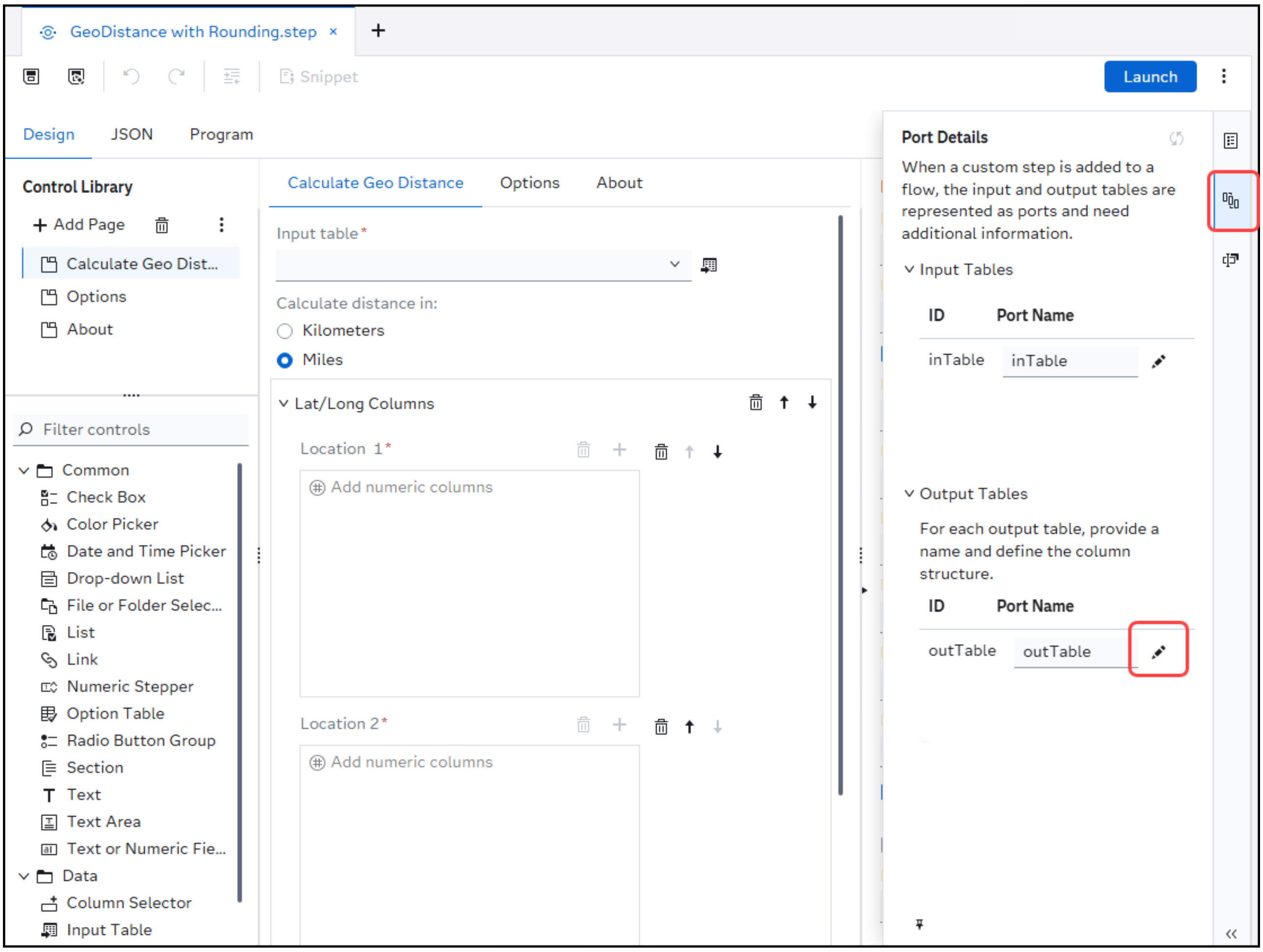


> ✎ The *ID* name for the control is **newColumn**.

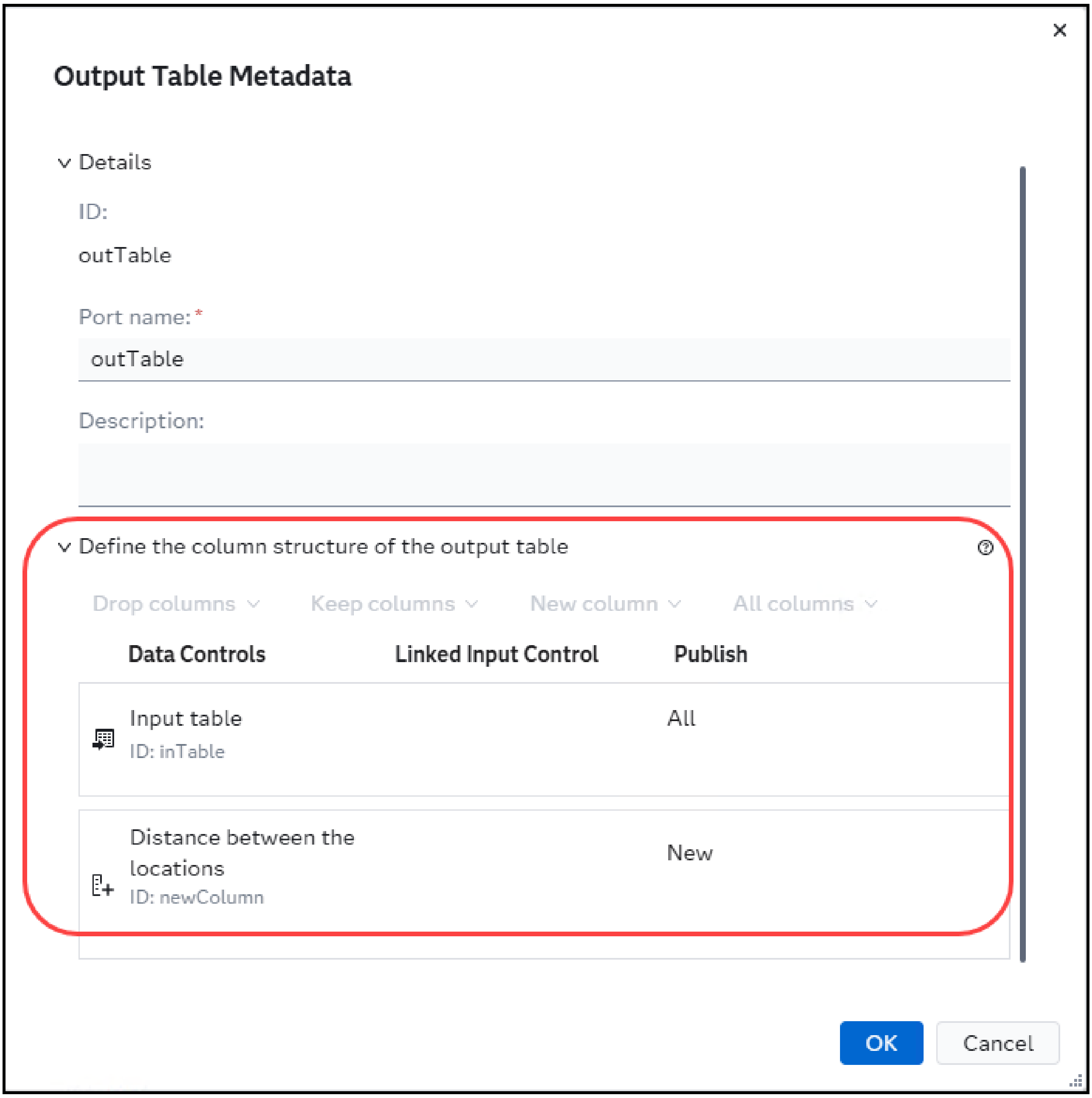
9. Use Port Details to Control the Output Columns

1. Click  to view the port details for the custom step.

2. Select  in the *Output Tables* section to view the output table structure for the custom step.



3. All the columns from the input table are part of the output in addition to the new column *Distance between the locations*.

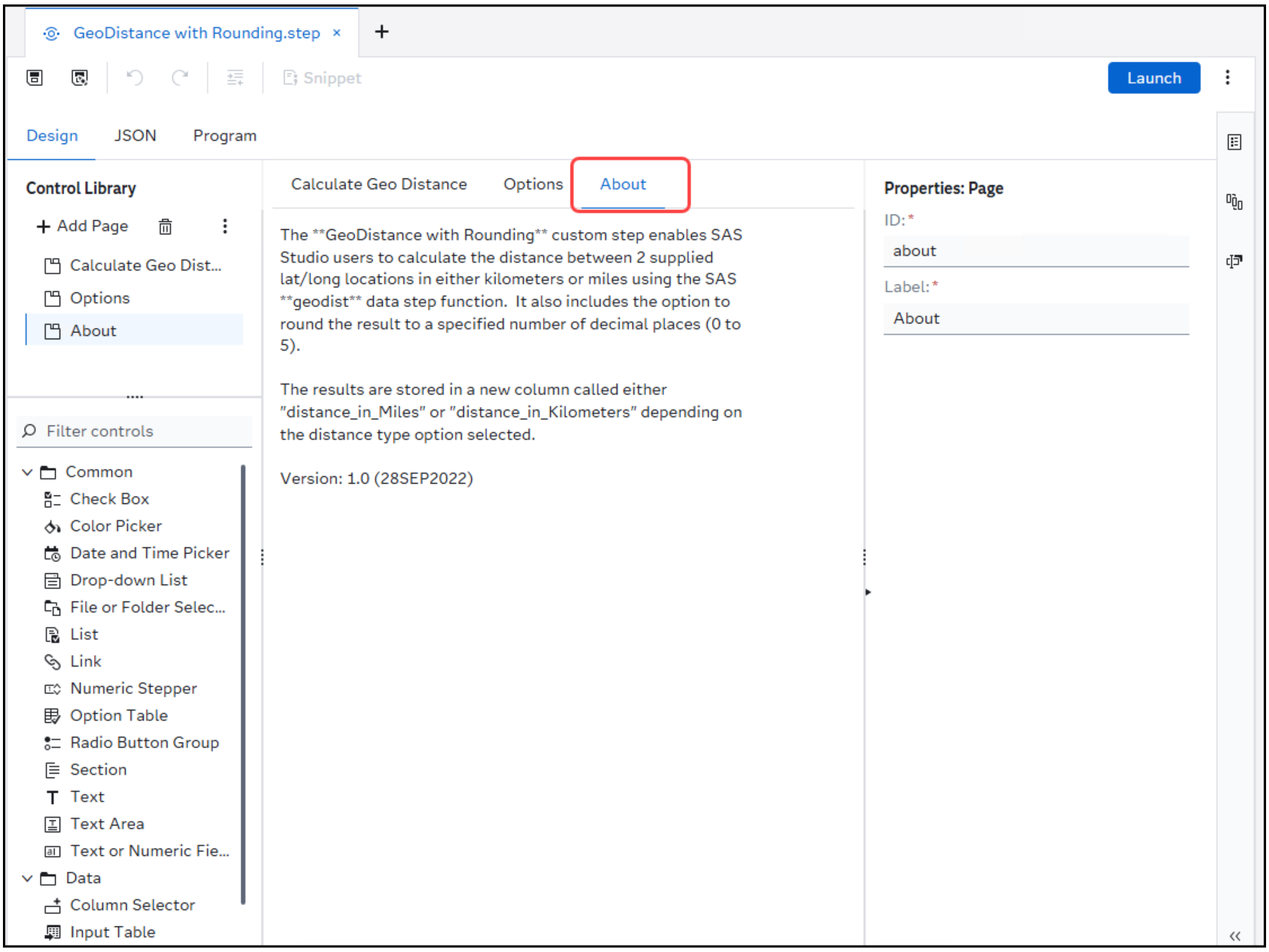


> ✎ If you decide to drop columns from the output table, then you should also explicitly drop the columns in the Program code as well.

4. Click **Cancel** to close the *Output Table Metadata* dialog without making any changes.

10. Create an **About** tab for the Custom Step

1. Click the **About** page tab for the custom step to view its properties. It is a best practice to include a *Text* control on this page with information about what the step does and its version information on this tab.



> ✎ To create a new page for an *About* tab, select **+ Add Page** in the **Control Library** section on the *Design* tab for the custom step.

2. Click **x** to close the *GeoDistance with Rounding.step* tab.

Exercise Completed

You have completed the exercise for **Tips & Tricks for Building a Custom Step with SAS Viya**.

THANK YOU FOR ATTENDING THIS SESSION!
PLEASE COMPLETE THE EVALUATION!

Custom Steps

Resources

Training Course

- Developing Custom Steps with SAS® Studio Analyst

Documentation

- SAS Help Center: SAS Studio: Working with Custom Steps

SAS Community Articles

- Search - SAS Support Communities for Custom Steps

Quick Start Video

- Create Custom Steps with SAS Studio

YouTube Video Series

- Creating a Custom Step in SAS Studio

GitHub Repository of Custom Steps

- https://github.com/sassoftware/sas-studio-custom-steps

sas innovate 2026

Training Course:

- Developing Custom Steps with SAS® Studio Analyst: https://learn.sas.com/course/view.php?id=220

Documentation:

- SAS Help Center: SAS Studio: Working with Custom Steps: https://go.documentation.sas.com/doc/en/sasstudiocdc/default/webeditorcdc/webeditorsteps/titlepage.htm

SAS Community Articles:

- Search - SAS Support Communities for Custom Steps: https://communities.sas.com/t5/forums/searchpage/tab/message?q=Custom%20Steps&noSynonym=false&collapse_discussion=true

Quick Start Video:

- Create Custom Steps with SAS Studio: https://video.sas.com/detail/video/6347011003112/create-custom-steps-with-sas-studio

YouTube Video Series:

- Creating a Custom Step in SAS Studio: https://youtube.com/playlist?list=PLVV6eZFA22QyLhfnKKW3k08XVx1K0ZCJ4&si=htu2O3BmS05oo9aD

GitHub Repository of Custom Steps:

- https://github.com/sassoftware/sas-studio-custom-steps

Thanks for attending this Hands-on Workshop session!

Please complete the evaluation for the session.

Mary Kathryn Queen
marykathryn.queen@sas.com

sas innovate 2026